

ActewAGL Distribution

2010/11 Pricing Proposal to the Australian Energy
Regulator April 2010



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ActewAGL Distribution Pricing Proposal 2010/11

April 2010

ActewAGL

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Overview

ActewAGL Distribution offers customers a range of network tariff options across 3 tariff classes – residential, commercial low voltage and high voltage. Customers are able to choose the tariff which best suits their needs, subject to some eligibility requirements as set out in this proposal.

ActewAGL Distribution proposes a number minor changes to tariff definitions and eligibility requirements in 2010/11. The price used in the renewable generation tariff for netting off and purchasing exports has been clarified for customers who may be on time-of-use tariffs. The off-peak (1) night network tariff has been re-opened and extended to include a range of new applications including recharging electric vehicles and compressing natural gas for CNG vehicles. Customers on the residential time-of-use tariff are eligible to use the off-peak tariffs. The exemption for public holidays has been removed from the definition of a weekday. In practice, this change will come into effect gradually as meters are replaced or reprogrammed.

As noted by the Australian Energy Regulator (AER) in the *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision*, over the course of the 2009-14 regulatory period ActewAGL Distribution will significantly increase investment on its network, which will result in higher prices for electricity consumers in the ACT. The AER also noted that increases in network charges are being driven by the introduction of the ACT feed-in tariff scheme. ActewAGL Distribution's proposed 2010/11 network tariffs are influenced also by significant increases in charges for transmission services, which must be passed on to customers in accordance with Chapter 11 of the *National Electricity Amendment Rules* (the *transitional Rules*).

ActewAGL Distribution's proposed distribution use of system (DUOS) charges for 2010/11 are 0.26 cents per kWh, or 5.4 per cent, higher on average than the charges for 2009/10. Transmission use of system (TUOS) charges are 0.23 cents per kWh, or 22.3 per cent, higher on average than the charges for 2009/10. Proposed network use of system (NUOS) charges (comprising both DUOS and TUOS) for 2010/11 are on average 0.49 c/kWh, or 8.4 per cent, higher than the average charges for 2009/10. Charges for miscellaneous services for 2010/11 are around 3 per cent higher than the 2009/10 charges.

Charges for metering services are subject to a separate regulatory cap. The proposed metering charges for 2010/11 are 4.6 per cent higher on average than the charges for 2009/10.

ActewAGL Distribution estimates that the proposed 2010/11 increases in network and metering charges will raise the electricity bill for an average residential customer consuming 7000 kWh on the residential basic network charge by \$0.72 per week (including GST), a real increase of 5.2 per cent (7.1 per cent nominal). For a commercial customer consuming 30 MWh per annum on the general network charge, the proposed network and metering price increases would raise their electricity bill by \$5.02 per week (including GST) or 7.0 per cent real (8.9 per cent nominal).

1. Introduction

1.1 Purpose and scope of the pricing proposal

ActewAGL Distribution has prepared this pricing proposal in accordance with the requirements in Chapter 11 of the *National Electricity Amendment Rules* (the transitional *Rules*). The pricing proposal provides the required information on the tariffs and charges to apply to ActewAGL Distribution's regulated distribution services from 1 July 2010 to 30 June 2011. A checklist of the transitional *Rules* requirements and where they are met in this proposal is provided in Attachment 1.

The pricing proposal relates to tariffs and charges to apply to ActewAGL Distribution's standard control services and alternative control services, as defined in clause 6.2.3C of the transitional *Rules*. ActewAGL Distribution's standard control services comprise distribution network use of system services and miscellaneous distribution services. ActewAGL Distribution's alternative control services comprise the provision and servicing of all manually read meters for consumers with loads less than 160 MWh per annum.

In accordance with the transitional *Rules*, separate regulatory control mechanisms apply to standard control and alternative control services. Therefore, separate price schedules must be determined. The combined standard control (network) and alternative control (metering) prices are also provided in this proposal.

The retail component of the delivered price of electricity to consumers accounts for around 60 per cent of the total price. It is fully open to competition, although there remains a regulated retail tariff offer for customers consuming less than 100 MWh per year that is set through a separate retail regulatory process.

1.2 Background

The Australian Energy Regulator (the AER) is responsible for the economic regulation of distribution services provided by ActewAGL Distribution. The AER has determined the maximum allowed average revenue (MAAR) for ActewAGL Distribution's standard control services and a revenue cap for alternative control (metering) services for the 2009-14 regulatory period.¹

This pricing proposal should be read in conjunction with the AER's final decision and ActewAGL Distribution's submissions and proposals (all published on the AER's website), as they set out in detail the basis of the costs that are reflected in the prices in this proposal. The AER summarises the outcomes from the regulatory review as follows:

¹ AER, *Final Decision, Australian Capital Territory distribution determination 2009-10 to 2013-14*, 28 April 2009.

Over the course of the next regulatory control period, ActewAGL will significantly increase investment on its network, which will result in higher prices for electricity consumers in the ACT. Higher prices will be largely driven by significant investment in four major capital projects. These projects include construction of two new zone substations, which are the first to be built in the ACT since 1994, augmentation of a third substation and construction of new assets to improve the security of electricity supply to the ACT. In addition, ActewAGL has already undertaken significant capital works to reinforce and replace a large number of unsafe poles, and these costs will now flow through to consumers. While consumers within the ACT will face higher charges as a result of this investment, they will also benefit from a more reliable and secure network.²

The AER also notes that in addition to higher capital investment, increases in network charges are also being driven by the introduction of the feed-in tariff scheme, which imposes a cost on ActewAGL Distribution.³

1.3 Structure of the pricing proposal

ActewAGL Distribution's tariff structure for standard control services is set out in chapter 2. The chapter includes details on the components and rationale for each tariff, an outline of ActewAGL Distribution's pricing strategy and how it relates to the pricing principles in the transitional *Rules*, and an explanation of the price setting process – that is, the process of moving from the regulated maximum allowed average revenue and a high level set of pricing principles to a full schedule of tariffs and charges for 2010/11.

The proposed network tariffs and charges for ActewAGL Distribution's standard control services for 2010/11 are presented in chapter 3. The chapter includes discussion of the proposed changes relative to 2009/10.

The structure and basis of ActewAGL Distribution's charges for alternative control (metering) services, the proposed charges for 2010/11 and the changes relative to 2009/10 are presented and explained in chapter 4.

Indicative estimates of the likely impacts of the pricing proposal on average customer electricity bills are provided in chapter 5.

² AER 2009, *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision*, April, p. viii

³ AER 2009, *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision*, April, p. viii

2. The structure and basis of ActewAGL Distribution's network tariffs

The transitional *Rules* (clause 6.18.2) require a description of the *tariff classes*⁴ and proposed tariffs that are to apply in 2010/11. For each tariff within a tariff class, the *charging parameters*⁵ and the elements of service to which they relate must also be set out in the pricing proposal.

2.1 Network tariff structure

ActewAGL Distribution's offers network tariffs in three tariff classes:

- Residential;
- Commercial low voltage (LV); and,
- High voltage (HV).

The transitional *Rules* say that tariff classes must be constituted with regard to the need to group customers together on an economically efficient basis and the need to avoid unnecessary transactions costs (clause 6.18.3(d)). ActewAGL Distribution meets this requirement by grouping customers according to type of connection (residential or commercial), and connection voltage (LV or HV). Customers within each class have similar load and connection characteristics. The relevant costs for each class can then be identified and reflected in the tariffs for each class.

Within each of these three tariff classes, ActewAGL Distribution has developed a suite of network tariffs that effectively meet the diverse needs of its customer base, encourage efficient use of the network and signal the costs of future network expansion. Residential customers are offered a choice of four network tariff options plus two controlled load off-peak options and an embedded renewable generation tariff option. Commercial LV customers are offered four main tariff options. Commercial customers on the general network charge also have access to the controlled load off-peak tariff options and the embedded renewable generation tariff option on a similar basis to customers in the residential class. Commercial HV customers are offered four tariff options. Customers are able to choose the option which best suits their needs, subject to the eligibility criteria set out in Tables 2.1 to 2.3 below.

The network tariffs comprise different combinations of the following *charging parameters*:

- Network access charges – apply per customer for residential consumers and per connection point for commercial customers. They involve a fixed daily charge and do not vary with electricity consumption or capacity;

⁴ A tariff class is defined as “a class of customers for one or more direct control services who are subject to a particular tariff or particular tariffs” in chapter 10 of the *National Electricity Rules*.

⁵ Charging parameters are defined as “the constituent elements of a tariff” in chapter 10 of the *National Electricity Rules*.

- Energy charges – apply to each unit of electricity consumed. The c/kWh rate may vary with the level of consumption (with higher rates applying above certain thresholds) and with the time-of-use (with lower rates applying at off-peak periods);
- Maximum demand charges – apply per connection point, for some commercial tariffs. They involve a charge per unit of maximum demand (in c/kVA/day). The maximum demand is the highest demand calculated over a 30-minute interval during the billing period.
- Capacity charges – apply on the same basis as the maximum demand charges, but are for the maximum demand recorded over the previous 12 months.

Network access charges relate to the connection services provided to customers. They are associated with the costs of the provision and use of connection assets for each tariff class, as well as customer related costs such as fault call centre costs.

Energy charges relate to the distribution services provided to customers. They are linked to the costs of the provision and use of distribution assets (other than connection assets), and also recover most of the common services costs. Higher energy rates at peak periods reflect higher costs. The higher energy rates beyond 330 kWh per day for the general network charge encourage larger consumers with a good load factor to move to demand or time-of-use network charges.

Maximum demand and capacity charges relate to the capacity provided to meet maximum demand and encourage consumers to manage their load on the network.

The allocation of costs to charging parameters is discussed further in section 2.4 below.

The tariffs and charging parameters for each tariff class are shown in the following tables (2.1 to 2.3). The tables include an explanation of the purpose of each tariff and the customers to which each tariff may apply.

2.1.1 Network tariffs for residential customers

ActewAGL Distribution's residential network tariff structure is shown in Table 2.1. The residential time-of-use (TOU), residential 5000 and residential with heat pump tariffs are refinements of the residential basic tariff to reflect customer load profiles. The residential TOU tariff provides an opportunity and an incentive for customers with the necessary metering capability to respond to price signals at different times of the day and so reduce their electricity bill in line with the costs they impose on the network. The residential 5000 and residential with heat pump tariffs involve a higher connection charge and an inclining block structure with a higher energy charge (per kWh) applying above certain thresholds. These tariffs more accurately tailor costs to the load profile of these customers. The off-peak tariff options can be used in conjunction with the residential basic tariff and the residential time-of-use network charge.

Table 2.1 Network tariff structure – residential

Tariff	Charging parameters	Explanation
Residential basic	Network access charge (c/day/customer)	The residential basic network tariff is available to installations

network	Energy charge (c/kWh)	at private dwellings, excluding serviced apartments, but including: • Living quarters for members and staff of religious orders; • Living quarters on farms; • Charitable homes; • Retirement villages; • Residential sections of nursing homes and hospitals; • Churches, buildings or premises which are primarily used for public worship; and • Approved caravan sites. The energy charge does not vary with the level of consumption nor the time of day. However, customers on this tariff are also eligible for the off-peak tariffs.
Residential time-of-use (TOU) network ⁶	Network access charge (c/day/customer) Energy at max times, ie 7am to 9am and 5pm to 8pm every day (c/kWh) Energy at mid times, ie 9am to 5pm and 8pm to 10pm every day (c/kWh) Energy at economy times, ie all other times (c/kWh)	This tariff is available to residential customers (as defined above) with a meter able to be read as a time-of-use meter. The energy charges relate to the supply of network services at various times. Higher rates apply at max or peak times to encourage users to shift their load to off-peak periods. Customers on this tariff are also eligible for the controlled load off-peak tariffs.
Residential 5000 network	Network access charge (c/day/customer) Energy for the first 60 kWh/day (c/kWh) Energy above 60 kWh/day (c/kWh)	This tariff is designed for residential customers who have large continuous (rather than time controlled) loads, such as electric hot water systems and consume over 5,000 kWh per annum. The energy charges relate to the supply of network services above and below certain volume thresholds. An inclining block structure applies (ie higher energy rates for the second block of energy). The lower energy rate is limited to consumption up to 60 kWh per day, reflecting a more typical domestic usage profile. This is sufficient to cover the energy requirements of many residential consumers.
Residential with heat pump	Network access charge (c/day/customer) Energy for the first 165 kWh/day (c/kWh) Energy above 165 kWh (c/kWh)	This tariff is only available to residential customers with a reverse cycle air conditioner. An inclining block structure applies (ie higher energy rates for the second block of energy). The lower energy rate is set to recover the incremental cost of energy load on the network as a demand management tool to lower winter peak loads and improve utilisation of the network in summer and so improve overall network utilisation.
Off-peak (1) night network	Energy at controlled times, ie between 10 pm and 7 am (c/kWh)	The off-peak (1) night charge is available only to customers utilising a controlled load element, and taking all other energy at residential basic network, residential time-off-use or general network rates. It is not available where the customer's meter is read as an interval meter. The off-peak (1) night charge is applicable to permanent: heat (or cold) storage; electric vehicle recharge; and CNG vehicle gas compression installations. The design and rating must be acceptable to ActewAGL. The installation must use most energy during the restricted times

⁶ All times for metering are Eastern Standard Time.

		but may be boosted at the principal charge, or charges, at other times. The off-peak (1) night network energy charge relates to supply of network services at controlled times, for 6 to 8 hours per day between the hours of 10 pm and 7 am.
Off-peak (3) day and night network	Energy at controlled times, ie between 10 pm and 7 am and 9 am and 5 pm (c/kWh)	Available only to customers utilising a controlled load element, and taking all other energy at residential basic network, residential time-off-use or general network rates. This charge is not available where the customer's meter is read as an interval meter. This charge is applicable to permanent: heat (or cold) storage installations. The design and rating must be acceptable to ActewAGL. The off-peak (3) day and night network energy rate applies to power supplied for up to 13 hours per day between 10 pm and 7 am and again between 9 am and 5 pm.
Renewable generation	Energy charges (c/kWh)	This tariff applies to customers with grid connected solar or wind energy generation systems. Different arrangements apply to customers participating in the ACT feed-in tariff scheme.

2.1.2 Network tariffs for low voltage commercial customers

ActewAGL Distribution sets different tariffs for low voltage commercial and high voltage commercial customers recognising the different costs associated with supplying each group. Within the commercial LV tariff class a range of tariff options has been developed to meet the diverse needs of commercial customers and to accommodate their differing load profiles and ability to respond to price signals. Of the four main options offered to commercial LV customers all but the general network tariff involve time-of-use charges. The general tariff does however involve an inclining block tariff structure with higher energy charges (c/kWh) applying above certain thresholds. Also, the off-peak (controlled load) tariffs are available to customers on the general network tariff.

Two of the commercial LV options involve capacity and/or maximum demand charges, in conjunction with time-of-use charges.⁷ Customers that are to improve their load factor⁸ have an incentive to choose a tariff with a demand or capacity charge, and reduce their energy bills. Customers on the general and general time-of-use network charge can move to the demand charges and lower their network costs if they have a sufficiently large load (for the cost savings to offset the higher cost of interval metering) and a reasonable load profile.

Table 2.2 Network tariff structure - commercial low voltage

Tariff	Charging parameters	Explanation
General network	Network access charge (c/day/customer) Energy for the first 330 kWh/day (c/kWh) Energy above 330 kWh/day (c/kWh)	This tariff is available to all customers. The tariff is most suitable for small commercial consumers operating in regular business hours or larger customers with poorer load factors (peaky loads).

⁷ A third option, the kW demand network charge is to be discontinued from 1 July 2009. There are no customers using this network charge.

⁸ The load factor is the actual energy consumed relative to the energy the customer would have consumed if it had consumed at its maximum rate of consumption for the full period.

		This tariff may be used in conjunction with the off-peak tariffs.
General TOU network	Network access charge (c/day/customer) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This tariff is particularly suitable for small commercial customers with discretionary or relatively large off-peak loads such as bakers, freezer installations and irrigators. The energy charges relate to supply of network services at different times.
LV TOU kVA demand network	Network access charge (c/day/connection point). Maximum demand (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This tariff is appropriate for customers with an average or stable commercial load. The maximum demand charge is designed to encourage consumers to manage their demand upon the network. The energy charges relate to supply of energy at different times, with lower rates in off-peak times reflecting the availability of capacity and encouraging consumers to shift their load from peak to off-peak times to utilize the available capacity.
LV TOU capacity network	Network access charge (c/day/connection point) Maximum demand (in billing period) (c/kVA/day) Capacity (max demand in last year) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This tariff is intended for low voltage customers with embedded generation but it may be an attractive option for other low voltage customers with seasonally stable loads. The tariff provides an incentive for customers with embedded generation to manage their output and their down-times (eg for servicing) so as to minimise their demand on the network.
Streetlighting	Network access charge (c/day/customer) Energy at any time (c/kWh)	This tariff applies to the night-time lighting of streets and public ways and places.
Small unmetered loads	Network access charge (c/day/customer) Energy at any time (c/kWh)	This tariff applies to eligible installations as determined by ActewAGL, including: • telephone boxes • telecommunication devices • other, as determined by the National Metrology Coordinator. Energy charges are calculated based on the assessed rating of the load and the charge period.
Internal (Obsolete)	Network access charge (c/day/customer) Energy at any time (c/kWh)	This tariff applies only to ActewAGL Distribution sites and fully recovers the economic costs of supply. It is now to be set at the rate of the first step in the general network tariff.

- Business times are between 7 am and 5 pm Eastern Standard Time on weekdays. Evening times are between 5 pm and 10 am Eastern Standard Time on weekdays. Off-peak times are all other times.

2.1.3 Network tariffs for high voltage customers

To qualify for the high voltage demand network charges, consumers must take their energy at high voltage (nominal voltage not less than 11 kV) and make a capital contribution towards their connection assets and transformers. High voltage consumers have the option of owning and operating their own high voltage assets. However, no consumers have selected this

option to date. Some customers have aggregated their load, incorporating part of ActewAGL Distribution's low voltage network to become a high voltage customer. A separate high voltage network charge is available for such customers. Customers taking their energy at high voltage also have the option of selecting the network tariffs available to low voltage consumers. For example, a high voltage customer with a poor load factor could select the general or the general time-of-use network charge.

Table 2.3 Network tariff structure - high voltage

<i>Tariff</i>	<i>Charging parameters</i>	<i>Explanation</i>
HV TOU Demand Network (111)	Network access charge (c/day/connection point) Max demand (in billing period) (c/kVA/day) Capacity (max demand in past year) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This tariff is appropriate for large customers taking supply at high voltage with a low voltage network owned and maintained by ActewAGL. The energy charges relate to supply of network services at different times, with lower rates in off-peak times encouraging customers to increase their utilisation of the network in off-peak periods. The demand charge is applied to the maximum demand in the billing period while the capacity charge is applied to the maximum demand in the previous 12 months. The capacity charge encourages the consumer to monitor and manage their peak demand over the year while the demand charge continues to encourage consumers to manage their demand requirements each month.
HV TOU Demand Network – Customer HV (112)	Network access charge (c/day/connection point) Max demand (in billing period) (c/kVA/day) Capacity (max demand in past year) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This tariff is appropriate for large customers taking supply at high voltage with a low voltage network owned and maintained by ActewAGL, where the customer owns and is responsible for their high voltage assets (including transformers and switch gear). No customers have selected this tariff.
<i>Tariff</i>	<i>Charging parameters</i>	<i>Explanation</i>
HV TOU Demand Network – Customer LV (121)	Network access charge (c/day/connection point) Max demand (in billing period) (c/kVA/day) Capacity (max demand in past year) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This network charge is appropriate for large customers taking supply at high voltage where the customer owns and is fully responsible for their own low voltage network. The network access charge relates to the connection services provided to the customer including provision of the current transformer necessary to meter these large loads. The capacity charge encourages the consumer to monitor and manage their peak demand over the year while the demand charge continues to encourage consumers to manage their capacity requirements each month. The energy charges relate to supply of network services at different times, with lower rates in off-peak times reflecting the relatively low costs of off-peak supply, and thereby providing incentives for customers to switch their utilisation of the network to off-peak periods.

HV TOU Demand Network – Customer HV and LV (122)	Network access charge (c/day/connection point) Max demand (in billing period) (c/kVA/day) Capacity (max demand in past year) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This network charge is appropriate for large customers taking supply at high voltage where the customer owns and is fully responsible for their own low voltage network and where the customer owns and is responsible for their high voltage assets (including transformers and switch gear). No customers have selected this tariff.
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* Business times are between 7 am and 5 pm Eastern Standard Time on weekdays. Evening times are between 5 pm and 10 am Eastern Standard Time on weekdays. Off-peak times are all other times.

2.1.4 Miscellaneous charges and fees

In addition to the network tariffs set out above, ActewAGL Distribution offers a range of miscellaneous services with charges for each service. The services are part of ActewAGL Distribution's standard control services, as set out in Appendix D of the AER's 2009-14 final decision. The charges are regulated through the maximum allowed average revenue cap applying to standard control services. The structure of charges varies, depending on the type of service. Some services are charged on a per visit basis, others per installation or per test. The charges for miscellaneous services are set on a cost reflective basis. For example, separate rates apply for temporary connections depending on whether they relate to an overhead connection, standard underground or free-standing underground, as these will involve different costs. Standard control miscellaneous services fees and charges are discussed further in chapter 3.

ActewAGL Distribution also provides some miscellaneous services as part of its alternative control (metering) services. These include meter tests, installation of interval meters at a customer's request, and special meter reads. These are covered in chapter 4.

2.2 Pricing strategy

ActewAGL Distribution has developed and refined the network tariff structure over time, guided by its pricing strategy. The strategy involves:

- Setting prices to signal to customers the economic costs of providing distribution services;
- Providing customers with a choice of flexible and innovative tariffs that best meet their needs;
- Providing incentives and opportunities for demand management;
- Ensuring that tariffs are set to recover costs in a way that encourages efficient use of the network and signals to customers the cost of network expansion; and,
- Offering customers a clear and simple tariff structure, noting the need to take account of the ability of different customer groups to respond to price signals and the need to keep transactions costs low.

ActewAGL Distribution's pricing strategy has resulted in the development of some innovative tariffs and significant customer responses in recent years. For example, in line with the strategies of setting cost reflective prices and providing opportunities and incentives for demand management, ActewAGL Distribution has gradually introduced several time-of-use charging options for both commercial and residential customers. More than 50 per cent of the total load in the ACT is now subject to time-of-use or controlled load (off-peak) charges. For the non-residential sector, 80 per cent of the load is on time-of-use or controlled load tariffs.

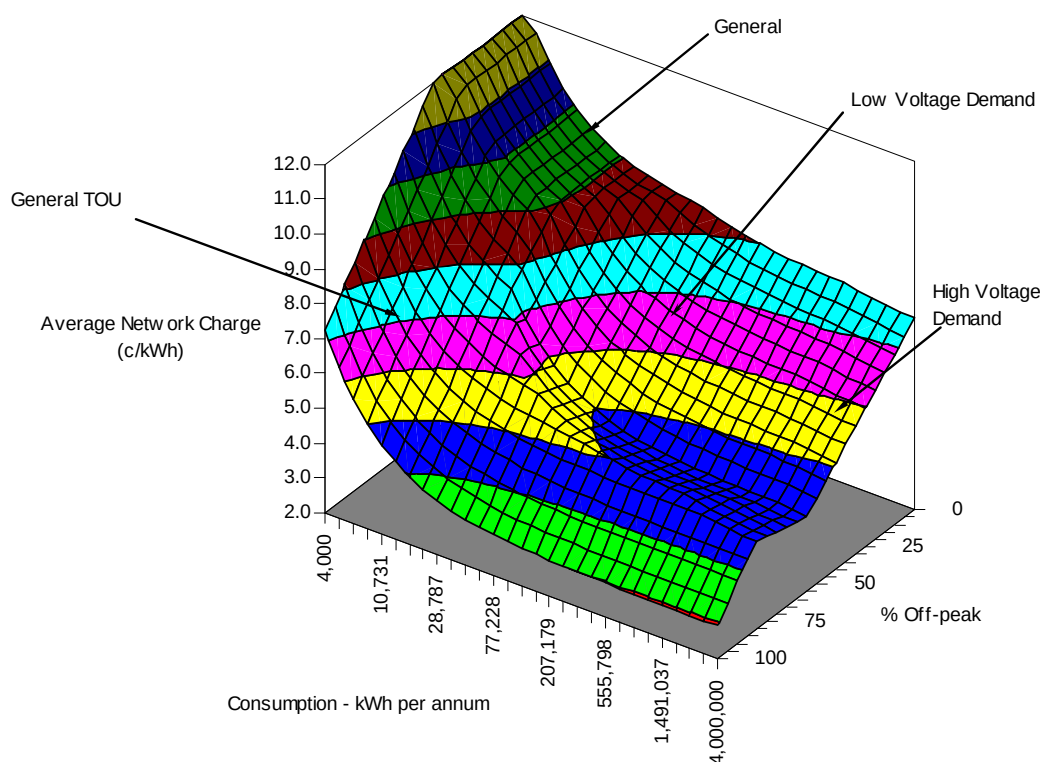
The introduction of maximum demand charges in several commercial tariff options has further strengthened price signals to customers, provided incentives to use the network more efficiently and resulted in significant customer responses. The maximum demand charges signal to customers the relatively high cost of providing capacity to meet demand and provide incentives to customers to improve both their load factor (that is, spread their load more evenly) and power factor (which allows the existing network to deliver more energy). Between 1999/00 and 2008/09 customers on the low voltage demand network tariff improved their load factor and therefore their utilisation of the network by 10 per cent, increasing the average energy consumed relative to the average of their monthly maximum demand from 40.1 per cent to 44.2 per cent. Over the same period, high voltage customers increased their load factor, and therefore their utilisation of the network, from 54.2 per cent to 58.8 per cent, an improvement of 8.5 per cent.

These price signals have been effective demand management tools and have allowed ActewAGL Distribution to keep network augmentation costs to a minimum. For example, no new zone substations have been built since 1994.

The options available for commercial customers and the incentives created are represented in Figure 1 which shows the main network tariffs available to commercial customers, including time-of-use and demand tariffs. Sophisticated modelling ensures that the tariffs are synchronised and that the costs of supplying particular loads are appropriately priced. ActewAGL Distribution is able to offer a suite of commercial tariffs and be indifferent to the customers' choice.

The graph plots the average minimum network charges over a range of consumption levels and a range of consumption profiles from all peak time (business and evening) to all off-peak for commercial customers. It shows how the general network charge suits small commercial customers with relatively large peak time loads while those with high off-peak loads can be rewarded by selecting the general time-of-use network tariff. The graph also demonstrates how the low voltage and high voltage demand tariffs align with the non-demand tariffs. Larger customers with good load profiles are able to choose a demand based tariff option and reduce overall network supply costs and their average network charges.

Figure 1 Average network charges for commercial customers



2.3 Consistency with the transitional Rules pricing principles

The transitional *Rules* include pricing principles to be applied when setting tariffs (clause 6.18.5). These principles recognise the complexities which arise in electricity network pricing because of the nature of electricity distribution costs, in particular the large fixed costs associated with providing capacity to meet peak demand, the high degree of shared or joint costs, and costs which vary according to the time of day and season and voltage of supply. There is no unique set of efficient and equitable prices, but rather a range within which prices should fall, taking account of the need to raise sufficient revenue to cover costs, the need to signal to customers the costs of expanding capacity to meet future demand, and the need to take account of the scope for customers to respond to price signals, bearing in mind the administrative and technical costs of doing so.

Clause 6.18.5 (a) of the transitional *Rules* requires that for each tariff class the revenue must lie between an upper bound of the stand alone costs and a lower bound of avoidable cost. The stand alone cost for any group of customers is the cost that would be incurred if only that customer group were supplied. Any costs that would otherwise be shared with other customer groups would have to be fully attributed to the stand alone customers. The stand alone cost is

effectively the cost of replicating or bypassing the infrastructure.⁹ The avoidable cost for any group of customers is the cost that would be avoided if the customers were removed from the network.

The purpose of the stand alone and avoidable cost test is to ensure that there are no cross subsidies between tariff classes. It is also designed to discourage inefficient bypass of the network (which may occur if prices are above the stand alone cost).¹⁰ If any tariff class, or group of customers, is paying less than the lower bound of the avoidable cost, or the costs that arise directly from its use of the network, then it is receiving a subsidy. If any tariff class is paying more than the upper bound of the stand alone cost, or the cost if it alone was supplied, then it subsidising other users. If the revenues from each group lie between the bounds of avoidable cost and stand alone cost then each group is making some contribution to covering the joint or shared costs of supply. However, no group is receiving or paying an economic subsidy.

Clause 6.18.5 (b) says that each tariff and charging parameter must take account of the long run marginal cost of the service (or element of the service) to which it relates, while also taking account of transactions costs and whether customers are able or likely to respond to price signals. The purpose of the long run marginal cost requirement is to ensure that prices signal to customers the forward-looking costs of meeting additional demand and the savings from reduced demand.

Clause 6.18.5 (c) says that if, as a result of the operation of 6.18.5 (b), expected revenue may not be covered, tariffs must be adjusted in a way that results in minimum distortion to efficient patterns of consumption.

ActewAGL Distribution's pricing strategy is consistent with each of the pricing principles in clause 6.18.5 of the transitional *Rules*. As far as possible, costs are allocated to the customer group or groups which are directly responsible for the costs. For example, the costs associated with providing low voltage connection assets are directly attributed to low voltage customers, but not high voltage customers. As a result, the revenue from each tariff class recovers at least the avoidable cost (or directly attributed cost) associated with providing the service to that class. Where additional costs must be recovered, each class pays a share, but none pays more than its stand alone cost. In this way, the requirement of providing subsidy-free prices is met. This is demonstrated in section 2.3.1 below, where ActewAGL Distribution's proposed revenue from each tariff class is shown to lie between the relevant bounds of avoidable and stand alone costs.

The principle of signalling to users the long run marginal cost is also taken into account in ActewAGL Distribution's approach to pricing. The long run marginal cost is the additional cost of meeting additional future demand. It includes the capital costs associated with any increment in capacity required to meet the additional demand, as well as the additional operating and maintenance costs. At times when demand is at or near capacity levels, prices

⁹ NERA 2006, *Distribution Pricing Rule Framework, Report prepared for the Network Policy Working Group of the Standing Committee of Officials (SCO) of the Ministerial Council on Energy (MCE)*, December, p. 22

¹⁰ NERA 2006, *Distribution Pricing Rule Framework*, December, p. 22

should signal the costs of expanding capacity rather than the short run marginal costs which do not include capital related costs.

ActewAGL Distribution's capacity and maximum demand charges provide signals to commercial customers about the cost of capacity expansion and, therefore, encourage these users to manage their maximum demands. The general time-of-use tariff does not incorporate a demand charge, but it does signal to customers the relatively high cost of providing capacity at peak times.

While residential tariffs do not have capacity or maximum demand charges, a time-of-use tariff is available and off-peak tariff options can be combined with other tariffs (including the residential basic tariff). The higher energy rates at peak periods signal to customers the cost of expanding capacity and, therefore, encourage consumers to shift their load to off-peak periods.

When setting tariffs to ensure that all costs are recovered, ActewAGL Distribution also aims to minimise the distortion from efficient patterns of consumption. This influences the balance between the network access charges (which do not vary with the level of consumption) and the energy charges (which do vary with the level of consumption and therefore more directly influence consumption patterns). For example, ActewAGL Distribution seeks to minimise the fixed charges in the general time-of-use and the low voltage demand tariff to ensure that small customers are not inefficiently constrained from adopting the tariffs with time-of-use and demand price signals. Also, for residential customers, ActewAGL Distribution has kept the fixed charge (including metering charge) for the residential time-of-use charge the same as for the basic residential charge to ensure there is not an inefficient barrier for consumers to move to a tariff that provides time-of-use price signals.

The practical application of these strategies and principles in the price setting process is described in section 2.4 below.

2.3.1 Calculating avoidable, stand alone and long run marginal costs

ActewAGL Distribution engaged economic consultants NERA to advise on the application of the pricing principles and to provide estimates of avoidable costs, stand alone costs and long run marginal costs. NERA undertook a detailed analysis of the available methods and developed and implemented a recommended approach to calculating avoidable, stand alone and long run marginal costs. NERA's report is provided as confidential Attachment 2.

NERA's estimates for avoidable and stand alone costs for each tariff class are shown in Table 2.4. The results in the table show that ActewAGL Distribution's proposed average 2010/11 DUOS revenue for each tariff class lies within the lower bound of avoidable cost and the upper bound of stand alone costs. The proposed tariffs therefore comply with the requirement in clause 6.18.5(a) of the transitional *Rules*.

Table 2.4 Avoidable and stand alone costs 2010/11 (\$'000)

Tariff class	Avoidable cost (‘000)	DUOS charges (‘000)	Stand alone cost (‘000)
Residential	\$8,641	\$56,223	\$145,527
Commercial low voltage	\$1,090	\$78,170	\$137,976
High voltage	\$56	\$13,432	\$136,942
Total		\$147,824	

ActewAGL Distribution notes that the avoidable cost estimates shown in Table 2.4 are significantly lower than those in the 2009/10 pricing proposal. This is primarily because NERA has adopted a different approach to the treatment of sunk costs in calculating avoidable costs, excluding them on the basis that the estimation of avoidable costs is a forward-looking exercise.¹¹

NERA's recommended approach also results in higher estimates of stand alone costs, compared with ActewAGL Distribution's 2009/10 estimates. While NERA notes that ideally the stand alone cost for each tariff class should be calculated on the basis of a hypothetical stand alone network, configured to supply only that class, it concludes that in practice this approach is time-consuming and subject to a high degree of uncertainty. NERA therefore recommends using ActewAGL Distribution's 2009/10 approach of measuring stand alone costs for each customer class as total cost less avoidable cost for the other classes. Lower estimates of avoidable cost (as discussed above), result in higher estimates of stand alone costs, compared with the estimates in the 2009/10 proposal. As the DUOS revenue is below the stand alone cost upper bound for each tariff class, ActewAGL Distribution's proposed tariffs comply with transitional *Rule* 6.18.5(a).

NERA's analysis of long run marginal costs highlights a range of practical and theoretical issues which need to be addressed in the calculation and interpretation of long run marginal cost, particularly where the distribution network may not be in a 'steady-state'.¹² ActewAGL Distribution considers that these matters should be carefully examined in the context of the AER's development of guidelines for the calculation of stand alone, avoidable and long run marginal costs (in accordance with clause 6.2.8 of the transitional *Rules*).

Following a detailed review of available methods and methods used in practice, NERA developed for ActewAGL Distribution a long run marginal cost model based on the average incremental cost (AIC) approach. Recognising that the long run marginal cost of network expansion varies depending on the network type (ie voltage level), NERA estimates the long run marginal cost for each network type then translates this into a cost estimate for each

¹¹ NERA 2010, *Analysis of ActewAGL's Electricity Distribution Services Costs, A report for ActewAGL*, commercial-in-confidence, April, p. 14

¹² For example, while the Turvey approach is generally considered to have the stronger theoretical basis, the Average Incremental Cost (AIC) method is more widely used as it is easier to apply. However, the AIC method has important shortcomings in that it is based on average capital costs associated with an increment in demand, not marginal, and the results are also sensitive to the demand forecasts used and the time frame for the analysis.

customer class on the basis of the contribution of each class to demand at times of network peak demand.¹³

NERA's estimates of long run marginal cost by network level, on a \$/kVA per year basis, are shown in Table 2.5. The estimates are expressed for each network level separately, with the low voltage estimate broken down to reflect the long run marginal cost of network assets used by low voltage commercial customers, low voltage residential customers and shared low voltage assets.

Table 2.5 Long run marginal costs per network level (\$/kVA pa)

<i>Network level</i>	<i>Long run marginal cost \$/kVA pa</i>
Low voltage residential	239.57
Low voltage commercial	29.76
Low voltage shared	4.74
High voltage	113.91

NERA's estimates of long run marginal cost for each tariff class are shown in Table 2.6. These long run marginal cost estimates represent the costs of meeting an additional kVA of demand for each of the three tariff classes.

- For customer located in the HV network, the long run marginal cost reflects the cost of augmenting the high voltage network, in order to meet that additional demand.
- For a low voltage commercial customer, the long run marginal cost reflects the cost of augmenting those areas of the low voltage network used exclusively by those customers, plus the costs of augmenting those areas of the low voltage network which are shared between low voltage commercial and residential customers and plus the costs of augmenting the high voltage network.
- Similarly, the long run marginal cost of meeting an increase in residential customer demand reflects the costs of augmenting those areas of the network used exclusively by residential customers plus those parts of the low voltage network which are shared between low voltage commercial and residential customers and plus the costs of augmenting the high voltage network.

As a result of applying this method of allocating costs, the long run marginal cost estimates for the low voltage customer classes are above the long run marginal cost estimates for the high voltage customer class.

¹³ NERA 2010, *Analysis of ActewAGL's Electricity Distribution Services Costs, A report for ActewAGL*, commercial-in-confidence, April, p. 11

Table 2.6 Long run marginal costs by tariff class (\$/kVA pa)

Tariff class	Long run marginal cost \$/kVA pa
Residential	367.06
Commercial low voltage	157.25
High voltage	113.91

NERA's long run marginal cost estimates are higher than the indicative long run marginal cost estimates provided in ActewAGL Distribution's 2009/10 pricing proposal. For the 2009/10 proposal ActewAGL Distribution calculated the average cost per kVA of a number of specific residential and commercial network expansion projects. NERA's approach is based on a broader set of information covering augmentation over the whole network and over a longer time period.¹⁴

ActewAGL Distribution notes that the long run marginal cost estimates are significantly influenced by a number of data limitations as well as features of the ACT demand and expenditure profiles in the current regulatory period, which mean that the distribution network is not in a steady-state, reflecting ActewAGL Distribution's stage in the investment cycle and the lumpy nature of network investment. Therefore care must be taken when interpreting the results and drawing implications for tariff policy.

For example, some of the expected future increase in electricity demand is associated with extensions of ActewAGL Distribution's existing distribution network to new sub-divisions. NERA explains that conceptually, the capital expenditure associated with extending the current network to meet additional demand in new areas can be distinguished from the capital expenditure associated with meeting increments in demand on the existing network. It is the latter which should be reflected in the estimate of long run marginal cost, as this provides the relevant behavioural signal to consumers. However in practice it is difficult to isolate incremental demand in existing network areas from demand forecasts for ActewAGL Distribution's overall network. As a result NERA calculated the long run marginal cost on the basis of total additional demand and total capital augmentation, including extension of the network to new areas.¹⁵

The unique circumstances in the current regulatory period, involving a significant increase in ActewAGL Distribution's forecast total capital augmentation costs, together with very low forecast growth rates for residential demand, result in proportionately higher estimates of long run marginal cost for residential customers. For these reasons, and consistent with the obligation to take long run marginal cost into account, the marginal cost estimate must be carefully interpreted and refined to reflect actual capacity availability in existing residential areas. Otherwise it would inadvertently provide a signal to existing customers that there were

¹⁴ NERA 2010, *Analysis of ActewAGL's Electricity Distribution Services Costs, A report for ActewAGL*, commercial-in-confidence, April, p. 19

¹⁵ NERA 2010, *Analysis of ActewAGL's Electricity Distribution Services Costs, A report for ActewAGL*, commercial-in-confidence, April, p. 20

capacity constraints when the existing residential network has sufficient capacity to meet demand.

ActewAGL Distribution also needs to take account of the best pricing mechanism to efficiently signal marginal costs via a mix of up-front changes (capital contributions) and recurrent charges.

ActewAGL Distribution intends to carefully assess options for addressing issues relating to the estimation of long run marginal cost using NERA's application of the AIC methodology, where the network is not in a steady-state.

2.4 The price setting process

The process of moving from the maximum allowed average revenue (MAAR) as set by the AER to a set of network tariffs involves the following steps:

1. Determine the maximum revenue to be recovered through distribution use of system (DUOS) charges. This requires some adjustments to the MAAR set by the AER in the final decision, as described in section 2.4.1 below.
2. Allocate the total revenue requirement to cost pools, taking account of the cost drivers (for example, whether the costs are demand related or customer related) and the type of assets involved (high voltage or low voltage). This cost allocation process is described in section 2.4.2 below.
3. Allocate the costs to tariff classes via a combination of network access charges, energy charges, demand charges and capacity charges. Costs are allocated according to the type of connection (residential, commercial, LV or HV) and the load profile. This process of setting the distribution use of system (DUOS) charges for each tariff class is described in section 2.4.3 below.
4. Allocate transmission use of system (TUOS) charges to tariff classes, to determine the total charges (DUOS + TUOS) to apply for each tariff class. The process of allocating TUOS charges is described in section 2.4.4 below.

2.4.1 Revenue to be recovered through DUOS charges

In accordance with the AER's final decision (chapter 4, section 4.5) and guideline for standard control services (appendix B, table 1), the maximum allowed average revenue for 2010/11 (in c/kWh) is converted to a total allowed revenue for 2010/11 by multiplying by 2009 throughput (in kWh). Approved cost pass-throughs are then deducted from this allowed revenue. Miscellaneous services charges are also deducted. They are part of the MAAR for standard control services but are set separately, as described above. The resulting value is the total maximum amount to be recovered through DUOS charges.

The relevant values for each of these components and the calculation of the DOUS cap for 2010/11 are provided in chapter 3 of this pricing proposal.

2.4.2 Allocation to cost pools

The total amount to be recovered in 2010/11 is allocated to three cost pools:

- Connection service costs
- Energy, demand and capacity service costs
- Common services costs

The main component of *connection service costs* is the cost of providing connection services to customers, that is, the cost associated with service mains (the wires that connect the customer's premises to the network).¹⁶ The cost pool also includes other costs that are related to the number of customers or connections, for example fault call centre costs.

The connection service costs are allocated across tariff classes using the proportion of connections in each tariff class relative to the total number of connections.

The main components of the *energy, demand and capacity service costs* are the costs of reticulation and providing capacity. The cost pool includes the costs associated with subtransmission assets (transformers, 132 kV substation assets) HV assets (eg high voltage distribution lines) and LV assets (LV substations and reticulation assets). The split between HV and LV costs is important as it ensures that LV and HV customers are allocated the appropriate costs. For example, HV customers without any ActewAGL LV reticulation are not required to pay for low voltage mains reticulation. This cost pool includes some shared costs which are demand-related, or related to the size of the network (for example training and apprenticeships costs), but cannot be attributed to certain voltages.

The energy, demand and capacity service costs are allocated across tariff classes on a cost reflective basis using energy consumed at different parts of the load cycle (business, evening and off-peak times), based on the method of intercepts.

Common services costs are shared across all users on the basis of energy consumption.

2.4.3 Allocating costs to tariff charging parameters

The cost of connection services and other customer related costs are generally recovered through the network access charges. Energy delivery, demand and capacity service costs are recovered through the energy, maximum demand and capacity charging parameters for those commercial tariffs where these components apply. Costs for common services are recovered in energy charges.

The allocation from cost pools to charging parameters is summarised in Table 2.7.

¹⁶ Metering assets and costs are regulated separately under the alternative control services control mechanism.

Table 2.7 Allocating service costs to tariff charging parameters

	Cost pools		
	Connection service costs	Energy, demand and capacity service costs	Common service costs
Residential			
Network access charge	Residential connection service costs and other customer related costs are recovered through network access charges.	For the residential 5000 and residential with heat pump network charges, some demand and capacity service costs are recovered via network access charges.	
Energy charges		Energy delivery, demand and capacity service costs for residential customers are generally recovered through residential energy charges.	Common service costs are recovered in energy charges.
Commercial LV			
Network access charge	Commercial connection service costs and other customer related costs are recovered through network access charges.		
Energy charges		Energy delivery service costs are recovered through the energy charge charges. Demand and capacity service costs are also recovered in energy charges where demand and capacity charging parameters are not applied.	Common service charges are recovered through the energy charges.
Capacity and maximum demand charges		Where demand and capacity charges apply, demand services and capacity services are recovered in demand and capacity charges.	
Commercial HV			
Network access charge	High voltage connection service costs and other customer related costs are recovered through network access charges		
Energy charges		Energy delivery costs are recovered through the energy charges.	Common service costs are recovered through the energy charges.
Capacity and maximum demand charges		Demand and capacity service costs are recovered through the demand and capacity charges.	

2.4.4 Allocating transmission use of system charges

In keeping with the strategy of setting prices to reflect the economic costs of providing network services, ActewAGL Distribution aims to preserve the economic signals present in the structure of TransGrid's transmission use of system (TUOS) charges when allocating the charges to tariff classes and components. For 2010/11, TransGrid recently advised (in a letter dated 12 April 2010) that it had decided to change its TUOS charging structure. As a result,

ActewAGL Distribution's charges have been quickly captured and incorporated so as to best preserve the intended price signals to customers. In particular, TransGrid has abolished the peak and shoulder energy charges and raised the flat energy charges. This has had implications particularly for ActewAGL Distribution's off-peak network energy charges.

Transgrid's transmission charges now comprise a fixed charge, an energy charge (with general and common service components) and a monthly maximum demand charge. The transmission exit charge is about 0.8 per cent of the total and varies according to the size of the connection. ActewAGL Distribution recovers this component of the transmission charge in its network energy charges or its demand and capacity charges (where possible), thereby ensuring that it is allocated according to the customer's utilisation of the transmission network. The transmission energy charges are also recovered in the network energy charges. For commercial tariffs which include a maximum demand charge, the demand component of the transmission charge is recovered in the maximum demand charge. In all other cases TransGrid's maximum demand charge is recovered in the network energy charges. In this way, ActewAGL's prices preserve the economic signals present in the structure of the TUOS charges where appropriate.

3. Proposed network tariffs for 2010/11

3.1 The revenue cap for standard control services

3.1.1 Base maximum allowable average revenue for standard control services

ActewAGL Distribution's standard control services prices are regulated using a maximum allowable average revenue (MAAR) cap. To calculate the maximum allowable average revenue for 2010/11, the following formula is used, as required by the AER's final decision (p. 18):

$$\text{Maximum allowable average revenue per kWh (MAAR}_t) = \text{MAAR}_{t-1} \times (1 + \text{CPI}_t) \times (1 - X_t)$$

where:

- MAAR_{t-1} is the maximum allowable average revenue per kWh in the previous year; and
- CPI means the Consumer Price Index, All Groups Index number for the weighted average of eight capital cities.

The CPI figure will be determined using the following formula:

$$\text{CPI}_{(t)} = \frac{\text{CPI}_{\text{March}(t-2)} + \text{CPI}_{\text{June}(t-2)} + \text{CPI}_{\text{Sept}(t-1)} + \text{CPI}_{\text{Dec}(t-1)}}{\text{CPI}_{\text{March}(t-3)} + \text{CPI}_{\text{June}(t-3)} + \text{CPI}_{\text{Sept}(t-2)} + \text{CPI}_{\text{Dec}(t-2)}} - 1$$

where:

- year t is the year for which tariffs are being set; and
- year t – 1 is the previous regulatory year.

The CPI adjustment figure for the year commencing 1 July 2010 is calculated as follows:

$$\text{CPI}_{(2010)} = \frac{\text{CPI}_{\text{March}(2009)} + \text{CPI}_{\text{June}(2009)} + \text{CPI}_{\text{Sept}(2009)} + \text{CPI}_{\text{Dec}(2009)}}{\text{CPI}_{\text{March}(2008)} + \text{CPI}_{\text{June}(2008)} + \text{CPI}_{\text{Sept}(2008)} + \text{CPI}_{\text{Dec}(2008)}} - 1$$

Therefore:

$$\text{CPI}_{(2010)} = \frac{166.2 + 167.0 + 168.6 + 169.5}{162.2 + 164.6 + 166.5 + 166.0} - 1$$

$$\text{CPI}_{(2010)} = \frac{671.3}{659.3} - 1$$

$$\text{CPI}_{(2010)} = 1.82\%$$

The maximum average revenue for 2009/10 calculated according to the AER's price determination for that year is \$0.04860 per kWh.¹⁷ The AER has determined the X factor for 2010/11 to be -4.0 per cent.¹⁸ The CPI of 1.82 per cent is applied to the maximum allowed average revenue (MAAR) for 2009/10 to obtain the maximum allowed average revenue in 2010/11. The calculations of the MAAR are shown in Table 3.1.

Table 3.1 Calculation of the Maximum Allowable Average Revenue 2010/11

	MAAR previous year	X Factor	Sum of CPI indices	CPI	MAAR
2008/09			631.8		\$0.04092
2009/10	\$0.04092	-13.82%	659.3	4.35%	\$0.04860
2010/11	\$0.04860	-4.00%	671.3	1.82%	\$0.05147

Note that while the CPI is shown as a percentage to 2 decimal places, the actual CPI figure applied the MAAR of the previous year is calculated based on the sum of the CPI indices for each year divided by the sum of the CPI indices for the previous year. Also, while the MAAR for 2008/09 is rounded to 5 decimal places, the MAAR figures for subsequent years are not rounded.

The maximum allowable average revenue cap is applied to the actual energy transported in the previous calendar year to establish a price cap for the following financial year. Therefore, the prices for 2010/11 are based upon energy transported in 2009. The actual energy transported in the 2009 calendar year was 2,892,502,567 kWh. When this is applied to the maximum allowable average revenue for 2010/11 of \$0.05147 per kWh, it results in the revenue cap for standard control services delivered in 2009 of \$148,866,962 which forms the regulated cap for 2010/11 prices.

3.1.2 Calculation of the revenue cap for DUOS

The calculation of the revenue to be recovered from 2010/11 distribution use of system (DUOS) charges is shown in Table 3.2.

Table 3.2 Calculation of the revenue cap for DUOS 2010/11*

Maximum allowable average revenue 2010/11 (\$/kWh)	A	\$0.05147
Energy sales ACT 2009 (kWh)	B	2,892,502,567
Standard control services revenue cap	$C = A \times B$	\$148,866,962
Approved cost pass-throughs	D	0
Total standard control services revenue cap	$E = C + D$	\$148,866,962
Less miscellaneous fees and charges	F	\$1,042,404
Distribution use of system revenue cap	$G = E - F$	\$147,824,558

* The MAAR shown in this table has been rounded to 5 decimal places. The calculations have been made without rounding.

¹⁷ ICRC 2004, *Final Decision – Investigation into prices for electricity distribution services in the ACT*, March, p. 132.

¹⁸ AER 2009, *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision*, April, p. 145.

3.2 Distribution use of system charges

ActewAGL Distribution proposes the distribution use of system (DUOS) prices for 2010/11 as shown in Table 3.3. These would have recovered \$147,824,492 on the actual customer, demand and energy quantities recorded in the 2009 calendar year. The table shows the tariff classes that are to apply in 2010/11 and the proposed tariffs for each tariff class. The charging parameters for each tariff are set out together with the service to which that charging parameter relates. All prices exclude GST unless otherwise stated.

To demonstrate compliance with the AER's control mechanism, ActewAGL Distribution is required to demonstrate that the sum of the standard control services revenue using the prices for the pricing year and the quantities for the previous calendar year divided by the quantity of energy in kWh transported over the previous calendar year is less than or equal to the MAAR for the pricing year.¹⁹

The sum of the distribution use of system charges and miscellaneous fees is \$148,866,896. When this is divided by the energy transported in the ACT of 2,892,502,567 kWh, it results in an average price of \$0.05147 per kWh. As this average price is equal to the MAAR, the proposed prices comply with the AER's final determination.

Table 3.3 Distribution use of system charges 2010/11

Tariff class & code	Tariff & element of service	Units	2009 kWh / cust no. / kVA	Proposed charges 2010/11	Notional revenue 2010/11
Residential tariffs					
010 Residential Basic Network					
	Network access	c/day/customer	127,088	14.01	\$6,498,817
	Energy at any time	c/kWh	914,260,655	4.701	\$42,979,393
015 Residential TOU Network					
	Network access	c/day/customer	6	14.01	\$325
	Energy at max times	c/kWh	13,861	7.103	\$985
	Energy at mid times	c/kWh	19,076	3.607	\$688
	Energy at economy times	c/kWh	15,177	2.530	\$384
020 Residential 5000 Network					
	Network access	c/day/customer	8,404	35.51	\$1,089,264
	Energy for the first 60 kWh per day	c/kWh	69,741,968	3.101	\$2,162,698
	Energy above 60 kWh per day	c/kWh	2,237,379	4.701	\$105,179
030 Residential with Heat Pump Network					
	Network access	c/day/customer	4,592	78.51	\$1,315,856
	Energy for the first 165 kWh per day	c/kWh	81,756,088	1.591	\$1,300,739
	Energy above 165 kWh per day	c/kWh	1,109,745	4.701	\$52,169
060 Off-Peak (1) Night Network					
	Energy at controlled times	c/kWh	17,050,086	0.302	\$51,491
070 Off-Peak (3) Day & Night Network					
	Energy at controlled times	c/kWh	94,777,739	0.702	\$665,340

¹⁹ AER 2009, *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision*, April, p. 18

Renewable Energy Generation

Energy generated & consumed	c/kWh	1,130,501	0.000	\$0
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Commercial low voltage tariffs

040 General Network

Network access	c/day/customer	12,250	27.66	\$1,236,725
Energy for the first 330 kWh per day	c/kWh	289,814,794	7.414	\$21,486,869
Energy above 330 kWh per day	c/kWh	24,081,159	10.134	\$2,440,385

135 Small Unmetered Loads Network

Network access	c/day/customer	23	20.30	\$1,680
Energy at any time	c/kWh	1,327,033	8.459	\$112,254

141 Internal Network (Obsolete)

Network access	c/day/customer	105	27.66	\$10,642
Energy at any time	c/kWh	1,542,848	7.414	\$114,387

080 Streetlighting Network

Network access	c/day/customer	20	28.00	\$2,078
Energy for night time lighting of streets public ways & places	c/kWh	39,690,736	4.471	\$1,774,573

090 General TOU Network

Network access	c/day/customer	280	27.66	\$28,316
Energy at business times	c/kWh	16,183,588	12.124	\$1,962,098
Energy at evening times	c/kWh	5,820,881	6.051	\$352,222
Energy at off-peak times	c/kWh	18,982,890	2.211	\$419,712

Low voltage time of use demand network

101 LV TOU kVA Demand Network

Network access	c/day/connection point	1,560	35.00	\$199,353
Maximum demand	c/kVA/day	238,241	32.99	\$28,687,397
Energy at business times	c/kWh	372,054,660	3.480	\$12,947,502
Energy at evening times	c/kWh	127,515,786	2.075	\$2,645,953
Energy at off-peak times	c/kWh	412,942,469	0.764	\$3,154,880

103 LV TOU Capacity Network

Network access	c/day/connection point	8	35.00	\$966
Maximum demand (in billing period)	c/kVA/day	2,683	15.18	\$148,665
Capacity (maximum demand in last year)	c/kVA/day	2,797	15.18	\$154,955
Energy at business times	c/kWh	4,982,570	3.480	\$173,393
Energy at evening times	c/kWh	2,412,452	2.075	\$50,058
Energy at off-peak times	c/kWh	8,441,272	0.764	\$64,491

High voltage tariffs

High voltage time of use demand network with ActewAGL low voltage network

111 HV TOU Demand Network

Network access	\$/day/connection point	1	13.00	\$7,098
Maximum demand (in billing period)	c/kVA/day	2,430	13.21	\$117,146
Capacity (maximum demand in last year)	c/kVA/day	3,470	13.21	\$167,310
Energy at business times	c/kWh	4,076,615	3.054	\$124,500
Energy at evening times	c/kWh	1,616,838	1.758	\$28,424
Energy at off-peak times	c/kWh	5,182,851	0.640	\$33,170

112 HV TOU Demand Network – Customer HV					
Network access	\$/day/connection point	0	13.00		\$0
Maximum demand (in billing period)	c/kVA/day	0	12.21		\$0
Capacity (maximum demand in last year)	c/kVA/day	0	12.21		\$0
Energy at business times	c/kWh	0	3.054		\$0
Energy at evening times	c/kWh	0	1.758		\$0
Energy at off-peak times	c/kWh	0	0.640		\$0
High voltage time of use demand network without ActewAGL low voltage network					
121 HV TOU Demand Network – Customer LV					
Network access	\$/day/connection point	21	13.00		\$97,682
Maximum demand (in billing period)	c/kVA/day	72,599	13.21		\$3,500,460
Capacity (maximum demand in last year)	c/kVA/day	85,096	13.21		\$4,103,024
Energy at business times	c/kWh	138,407,861	2.634		\$3,645,663
Energy at evening times	c/kWh	52,602,475	1.388		\$730,122
Energy at off-peak times	c/kWh	182,710,514	0.480		\$877,010
122 HV TOU Demand Network – Customer HV and LV					
Network access	\$/day/connection point	0	13.00		\$0
Maximum demand (in billing period)	c/kVA/day	0	12.21		\$0
Capacity (maximum demand in last year)	c/kVA/day	0	12.21		\$0
Energy at business times	c/kWh	0	2.634		\$0
Energy at evening times	c/kWh	0	1.388		\$0
Energy at off-peak times	c/kWh	0	0.480		\$0
Total					\$147,824,492
Total Customers					154,359
Total Energy Consumption					2,892,505,364

Table 3.4 sets out for each tariff class related to standard control services the expected weighted average DUOS revenue²⁰ for the relevant regulatory year and the current regulatory year, as required by clause 6.18.2(b)(4) of the transitional *Rules*.

Clause 6.18.6 of the transitional *Rules* applies a side constraint on the expected weighted average revenue to be raised from tariff classes. The side constraint is given by the formula $(1 + \text{CPI})(1 - X)(1 + 2\%)$. For 2010/11, the CPI is 1.82 per cent and the X factor is -4 per cent. Therefore, the side constraint is 8.01 per cent. As shown in Table 3.4, the price increase for all three tariff classes is within the side constraint.

Table 3.4 Weighted average DUOS revenue by tariff class

Tariff class	2009/10 (c/kWh)	2010/11 (c/kWh)	Change %
Residential	4.5	4.8	5.9%
Commercial Low Voltage	5.6	5.9	5.4%
High Voltage	3.4	3.5	3.5%
Average	4.8	5.1	5.4%

²⁰ This is the total DUOS revenue for each class divided by the total energy delivered to each class.

3.3 Miscellaneous services charges

As set out in the AER's guideline for standard control services, prices for miscellaneous services are regulated under the standard control services revenue cap. Like the proposed DUOS charges, the proposed charges for miscellaneous services for year "t" are checked for compliance with the revenue constraint for year "t" prices by multiplying the proposed prices for the coming year by actual sales in the previous calendar year, for each service.²¹

ActewAGL Distribution's proposed charges for miscellaneous services are shown in Table 3.5. These charges, when applied to units sold in 2009, would have recovered \$1,042,404.

Table 3.5 Miscellaneous fees and charges 2010/11

Code	Description	Units	Units 2009	Proposed prices 2010/11	Revenue with 2010/11 prices
For visit to re-energise or de-energise a site					
501	Business hours – re-energise a site	per visit	4,830	\$49.09	\$237,106
502	After hours – re-energise a site	per visit	1,297	\$105.45	\$136,769
503	Business hours – de-energise a site	per visit	2,584	\$43.64	\$112,766
505	De-energise a site for non-payment	per visit	653	\$81.82	\$53,428
508	Field visit only (De-energise a site for non-payment)	per visit	-	\$59.09	\$0
Temporary connections					
520	Overhead	per installation	109	\$350.00	\$38,101
521	Standard underground	per installation	1	\$508.18	\$508
522	Free-standing underground	per installation	11	\$616.36	\$6,902
Services					
530	Overhead: remove, reposition or disconnect service	per site visit	457	\$251.82	\$115,112
531	Underground: remove, reposition or disconnect service	per site visit	454	\$606.36	\$275,496
Upgrade service from single to three phase at customer's request where load does not justify three phase					
532	Overhead	per installation	0	\$325.45	\$0
533	Underground – service cable replacement not required	per installation	0	\$325.45	\$0
534	Underground – service cable replacement required	per installation	0	\$606.36	\$0

²¹ AER 2008, *Guideline on control mechanisms for direct control services for the ACT and NSW 2009 distribution determinations*, February, p. 5.

540	Rescheduled visit (applied where a revisit to a site is necessitated by obstructed access, non-compliance with the Service and Installation Rules or the client is not ready for the scheduled work.)				
		per visit	0	\$167.27	\$0
550	Issue of copies of electrical drawings	per sheet or electronic equivalent	141	\$13.09	\$1,842
560	De-energising wires (to allow safe approach, for example, for tree pruning, plant operation, oversize loads, construction activities)	per occurrence	159	\$404.55	\$64,374
570	Embedded generation - charges for small embedded generators (excluding renewable < 30KW)				
	for operation and maintenance of connection assets	per annum applied to the value of connection assets appreciated each year by CPI		2.00%	\$0
	for operation and maintenance of shared network assets (if applicable)	per annum applied to the value of shared assets allocated to the generator appreciated by CPI		2.00%	\$0
Total standard control miscellaneous services					\$1,042,404

3.4 Transmission use of system charges

The AER separately regulates transmission use of system (TUOS) charges. TUOS charges are not part of ActewAGL Distribution's regulated revenue requirement. Clause 6.18.7(a) of the transitional *Rules* allows ActewAGL Distribution to pass on to customers the charges to be incurred by ActewAGL Distribution for transmission use of system services. Clause 6.18.7(b) says that the amount to be passed on must not exceed the estimated amount of the TUOS charges for the relevant regulatory year adjusted for under or over recovery in the previous regulatory year. Clause 6.18.7(c) describes the method to be applied in determining the extent of under or over recovery.

To demonstrate compliance with clause 6.18.7 of the transitional *Rules* and the AER's final decision, the AER requires ActewAGL Distribution to maintain a transmission use of system (TUOS) overs and unders account. Clause 6.18.2(b)(7) requires ActewAGL Distribution to provide information on this account as part of the annual pricing proposal. The AER requires ActewAGL to provide details of its calculations in the format set out in Table E.1 of the final decision. Table 3.6 provides the required information.

Table 3.6 TUOS overs and unders account

	2008/09 Actual	2009/10 Estimate	2010/11 Forecast
Revenue from TUOS charges	20,376	30,305	36,593
Transmission charges to be paid to TNSPs	22,409	28,411	36,681
Avoided TUOS payments	0	0	0
Inter-DNSP payments	0	0	0
Total transmission related payments	22,409	28,411	36,681
Over (under) recovery for the financial year	-2,033	1,894	-89
Overs and unders account			
Annual rate of interest applicable to balances	9.31%	8.79%	8.79%
Semi-annual interest rate	4.55%	4.30%	4.30%
Opening Balance	355	-1,737	85
Interest on opening balance	33	-153	7
Over/under recovery for financial year	-2,033	1,894	-89
Interest on over/under recovery	-92	81	-4
Closing balance	-1,737	85	0

The forecast revenue from TUOS charges shown in Table 3.6 for 2010/11 is \$36,592,695, an increase of 20.8 per cent compared to forecast TUOS revenue for 2009/10. This amount has been calculated on the basis of TransGrid's published prices for 2010/11 and the forecast energy imported into the ACT and includes a refund of \$88,795 based on a forecast over-collection in 2009/10.

Based on TransGrid's prices, ActewAGL Distribution has forecast TUOS charges would be \$36,681,490 in 2010/11, an increase of 29.1 per cent. ActewAGL Distribution recovers TransGrid's TUOS charges from ACT consumers according to the energy they consume, and where possible, according to maximum demand in a month and over the year (where capacity charges apply). The cost allocations take into account the load profile of each customer class.

Only one set of TUOS prices are calculated, incorporating any adjustments to the total TUOS revenue requirement resulting from over or under recovery of TUOS charges in the previous regulatory year. The consumption profile used to calculate TUOS prices is the same 2009 consumption profile used to calculate DUOS prices.

As the energy transported in 2010/11 is forecast to be about 0.8 per cent higher²² than the actual energy transported in the 2009 calendar year, the amount of TUOS to be recovered in prices using the load in 2009 needs to be adjusted to \$36,297,686 or about 0.8 per cent

²² This is based on the forecasts adopted by the AER in the final decision – see chapter 7.

lower.²³ That amount has been used to calculate the TUOS prices in Table 3.7. The proposed TUOS prices would have recovered revenue of \$36,297,675 under the 2009 profile.

Table 3.7 Transmission use of system charges 2010/11

Code	Description	Unit	2009 kWh/ cust. no. /kVA	Proposed price 2010/11	Notional TUOS revenue
Residential tariffs					
010	Residential Basic Network				
	Network access charge	cents/day	127,088	0.00	\$0
	Energy consumption	cents/kWh	914,260,655	1.149	\$10,504,855
015	Residential TOU Network				
	Network access charge	cents/day	6	0.00	\$0
	Energy at max times	cents/kWh	13,861	1.327	\$184
	Energy at mid times	cents/kWh	19,076	1.143	\$218
	Energy at economy times	cents/kWh	15,177	1.000	\$152
020	Residential 5000 Network				
	Network access charge	cents/day	8,404	0.00	\$0
	Energy for the first 60 kWh per day	cents/kWh	69,741,968	1.149	\$801,335
	Energy above 60 kWh per day	cents/kWh	2,237,379	1.149	\$25,707
030	Residential with Heat Pump Network				
	Network access charge	cents/day	4,592	0.00	\$0
	Energy for the first 165 kWh per day	cents/kWh	81,756,088	1.149	\$939,377
	Energy above 165 kWh per day	cents/kWh	1,109,745	1.149	\$12,751
060	Off-Peak (1) Night Network				
	Energy consumption	cents/kWh	17,050,086	0.858	\$146,290
070	Off-Peak (3) Day & Night Network				
	Energy consumption	cents/kWh	94,777,739	0.858	\$813,193
	Renewable Energy Generation				
	Energy consumption	cents/kWh	1,130,501	0.000	\$0
Commercial low voltage tariffs					
040	General Network				
	Network access charge	cents/day	12,250	0.00	\$0
	Energy for the first 330 kWh per day	cents/kWh	289,814,794	1.616	\$4,683,407
	Energy above 330 kWh per day	cents/kWh	24,081,159	1.766	\$425,273
135	Small Unmetered Loads Network				
	Network access charge	cents/day	23	0.00	\$0
	Energy consumption	cents/kWh	1,327,033	1.331	\$17,663
141	Internal Network (Obsolete)				
	Network access charge	cents/day	105	0.00	\$0
	Energy consumption	cents/kWh	1,542,848	1.616	\$24,932
080	Streetlighting Network				

²³ AER, *Final decision, Control mechanisms for direct control services for the ACT and NSW 2009 distribution determinations*, February 2008, part 5.8.

	Network access charge	cents/day	20	0.00	\$0
	Energy consumption	cents/kWh	39,690,736	0.899	\$356,820
090	General TOU Network				
	Network access charge	cents/day	280	0.00	\$0
	Energy at business times	cents/kWh	16,183,588	1.876	\$303,604
	Energy at evening times	cents/kWh	5,820,881	1.269	\$73,867
	Energy at off-peak times	cents/kWh	18,982,890	0.869	\$164,961
Low voltage time of use demand network					
101	LV TOU kVA Demand Network				
	Network access per connection point	cents/day	1,560	0.00	\$0
	Maximum demand charge	c/kVA/day	238,241	4.61	\$4,008,757
	Energy at business times	cents/kWh	372,054,660	0.940	\$3,497,314
	Energy at evening times	cents/kWh	127,515,786	0.895	\$1,141,266
	Energy at off-peak times	cents/kWh	412,942,469	0.876	\$3,617,376
103	LV TOU Capacity Network				
	Network access per connection point	cents/day	8	0.00	\$0
	Maximum demand charge	c/kVA/day	2,683	2.12	\$20,762
	Capacity charge	c/kVA/day	2,797	2.12	\$21,641
	Energy at business times	cents/kWh	4,982,570	0.940	\$46,836
	Energy at evening times	cents/kWh	2,412,452	0.895	\$21,591
	Energy at off-peak times	cents/kWh	8,441,272	0.876	\$73,946
High voltage tariffs					
High voltage time of use demand network with ActewAGL low voltage network					
111	HV TOU Demand Network				
	Network access per connection point	\$/day	1	0.00	\$0
	Maximum demand charge	c/kVA/day	2,430	1.99	\$17,647
	Capacity charge	c/kVA/day	3,470	1.99	\$25,204
	Energy at business times	cents/kWh	4,076,615	0.896	\$36,526
	Energy at evening times	cents/kWh	1,616,838	0.872	\$14,099
	Energy at off-peak times	cents/kWh	5,182,851	0.860	\$44,573
112	HV TOU Demand Network – Customer HV				
	Network access per connection point	\$/day	0	0.00	\$0
	Maximum demand charge	c/kVA/day	0	1.99	\$0
	Capacity charge	c/kVA/day	0	1.99	\$0
	Energy at business times	cents/kWh	0	0.896	\$0
	Energy at evening times	cents/kWh	0	0.872	\$0
	Energy at off-peak times	cents/kWh	0	0.860	\$0
High voltage time of use demand network without ActewAGL low voltage network					
121	HV TOU Demand Network – Customer LV				
	Network access per connection point	\$/day	21	0.00	\$0
	Maximum demand charge	c/kVA/day	72,599	1.99	\$527,321
	Capacity charge	c/kVA/day	85,096	1.99	\$618,094
	Energy at business times	cents/kWh	138,407,861	0.896	\$1,240,134
	Energy at evening times	cents/kWh	52,602,475	0.872	\$458,694
	Energy at off-peak times	cents/kWh	182,710,514	0.860	\$1,571,310
122	HV TOU Demand Network – Customer HV and LV				
	Network access per connection point	\$/day	0	0.00	\$0

Maximum demand charge	c/kVA/day	0	1.99	\$0
Capacity charge	c/kVA/day	0	1.99	\$0
Energy at business times	cents/kWh	0	0.896	\$0
Energy at evening times	cents/kWh	0	0.872	\$0
Energy at off-peak times	cents/kWh	0	0.860	\$0
Total				\$36,297,682

3.5 Network use of system charges

ActewAGL Distribution's proposed network use of system (NUOS) charges for 2010/11 comprise the distribution use of system (DUOS) charges and transmission use of system (TUOS) charges. The proposed NUOS charges are shown in Table 3.8. All charges exclude GST.

Table 3.8 Network use of system charges 2010/11

Code	Description	Unit	Distribution Charges 2010/11	Transmission Charges 2010/11	Network Charges 2010/11
Residential tariffs					
010	Residential Basic Network				
	Network access charge	cents/day	14.01	0.00	14.01
	Energy consumption	cents/kWh	4.701	1.149	5.850
015	Residential TOU Network				
	Network access charge	cents/day	14.01	0.00	14.01
	Energy consumption at max times	cents/kWh	7.103	1.327	8.430
	Energy consumption at mid times	cents/kWh	3.607	1.143	4.750
	Energy consumption at economy times	cents/kWh	2.530	1.000	3.530
020	Residential 5000 Network				
	Network access charge	cents/day	35.51	0.00	35.51
	Energy consumption for the first 60 kWh per day	cents/kWh	3.101	1.149	4.250
	Energy consumption above 60 kWh per day	cents/kWh	4.701	1.149	5.850
030	Residential with Heat Pump Network				
	Network access charge	cents/day	78.51	0.00	78.51
	Energy consumption for the first 165 kWh per day	cents/kWh	1.591	1.149	2.740
	Energy consumption above 165 kWh per day	cents/kWh	4.701	1.149	5.850
060	Off-Peak (1) Night Network				
	Energy consumption	cents/kWh	0.302	0.858	1.160
070	Off-Peak (3) Day & Night Network				
	Energy consumption	cents/kWh	0.702	0.858	1.560
	Renewable Energy Generation				
	Energy consumption	cents/kWh	0.000	0.000	0.000
Commercial low voltage tariffs					
040	General Network				

	Network access charge	cents/day	27.66	0.00	27.66
	Energy consumption for the first 330 kWh per day	cents/kWh	7.414	1.616	9.030
	Energy consumption above 330 kWh per day	cents/kWh	10.134	1.766	11.900
135	Small Unmetered Loads Network				
	Network access charge	cents/day	20.30	0.00	20.30
	Energy consumption	cents/kWh	8.459	1.331	9.790
141	Internal Network (Obsolete)				
	Network access charge	cents/day	27.66	0.00	27.66
	Energy consumption	cents/kWh	7.414	1.616	9.030
080	Streetlighting Network				
	Network access charge	cents/day	28.00	0.00	28.00
	Energy consumption	cents/kWh	4.471	0.899	5.370
090	General TOU Network				
	Network access charge	cents/day	27.66	0.00	27.66
	Energy consumption at business times	cents/kWh	12.124	1.876	14.000
	Energy consumption at evening times	cents/kWh	6.051	1.269	7.320
	Energy consumption at off-peak times	cents/kWh	2.211	0.869	3.080
Low voltage time of use demand network					
101	LV TOU kVA Demand Network				
	Network access charge per connection point	cents/day	35.00	0.00	35.00
	Maximum demand charge	c/kVA/day	32.99	4.61	37.60
	Energy consumption at business times	cents/kWh	3.480	0.940	4.420
	Energy consumption at evening times	cents/kWh	2.075	0.895	2.970
	Energy consumption at off-peak times	cents/kWh	0.764	0.876	1.640
103	LV TOU Capacity Network				
	Network access charge per connection point	cents/day	35.00	0.00	35.00
	Maximum demand charge	c/kVA/day	15.18	2.12	17.30
	Capacity charge	c/kVA/day	15.18	2.12	17.30
	Energy consumption at business times	cents/kWh	3.480	0.940	4.420
	Energy consumption at evening times	cents/kWh	2.075	0.895	2.970
	Energy consumption at off-peak times	cents/kWh	0.764	0.876	1.640
High voltage tariffs					
High voltage time of use demand network with ActewAGL low voltage network					
111	HV TOU Demand Network				
	Network access charge per connection point	\$/day	13.00	0.00	13.00
	Maximum demand charge	c/kVA/day	13.21	1.99	15.20
	Capacity charge	c/kVA/day	13.21	1.99	15.20
	Energy consumption at business times	cents/kWh	3.054	0.896	3.950
	Energy consumption at evening times	cents/kWh	1.758	0.872	2.630
	Energy consumption at off-peak times	cents/kWh	0.640	0.860	1.500
112	HV TOU Demand Network – Customer HV				
	Network access charge per connection point	\$/day	13.00	0.00	13.00
	Maximum demand charge	c/kVA/day	12.21	1.99	14.20
	Capacity charge	c/kVA/day	12.21	1.99	14.20

Energy consumption at business times	cents/kWh	3.054	0.896	3.950
Energy consumption at evening times	cents/kWh	1.758	0.872	2.630
Energy consumption at off-peak times	cents/kWh	0.640	0.860	1.500
High voltage time of use demand network without ActewAGL low voltage network				
121 HV TOU Demand Network – Customer LV				
Network access charge per connection point	\$/day	13.00	0.00	13.00
Maximum demand charge	c/kVA/day	13.21	1.99	15.20
Capacity charge	c/kVA/day	13.21	1.99	15.20
Energy consumption at business times	cents/kWh	2.634	0.896	3.530
Energy consumption at evening times	cents/kWh	1.388	0.872	2.260
Energy consumption at off-peak times	cents/kWh	0.480	0.860	1.340
122 HV TOU Demand Network – Customer HV and LV				
Network access charge per connection point	\$/day	13.00	0.00	13.00
Maximum demand charge	c/kVA/day	12.21	1.99	14.20
Capacity charge	c/kVA/day	12.21	1.99	14.20
Energy consumption at business times	cents/kWh	2.634	0.896	3.530
Energy consumption at evening times	cents/kWh	1.388	0.872	2.260
Energy consumption at off-peak times	cents/kWh	0.480	0.860	1.340

3.6 Proposed changes to network tariffs

The transitional *Rules* (6.18.2(b)(8)) require an explanation of the nature and extent of changes from the previous regulatory year.

The proposed changes from 2009/10 to the residential network tariff structure are as follows:

- The “residential standard network charge” is renamed the “residential basic network charge”;
- The off-peak (1) night network charge is to be re-opened and extended to include recharging electric vehicles and compressing natural gas for CNG vehicles. The reference to off-peak 2 has been removed as the two off-peak tariffs now have the same price and application;
- The exemption for public holidays has been removed from the definition of a weekday. In practice, this change will come into effect gradually as meters are replaced or reprogrammed;
- The “energy consumption at controlled economy times” tariff rate offered with the Residential TOU Network charge is to be removed as customers on that tariff will now have access to both the off-peak network charges (no customers are affected);
- The internal tariff is now obsolete; and
- The buy-back rates for customers on stepped or time-of-use tariffs that are also on the “renewable generation tariff” are clarified. The new arrangements are as follows:

If a customer has a grid-connected renewable energy generator and the customer is not receiving the ACT feed-in tariff²⁴, the following arrangements apply:

- The customer will pay the published network charge for the gross amount of energy imported from ActewAGL Distribution's network.
- ActewAGL Distribution will pay to the customer's retailer an amount equal to the customer's network energy tariff rate on the gross amount of energy exported.
- If the customer is on a time-of-use tariff, the customer's network energy tariff rate for the purpose of paying the customer's retailer shall be the middle rate (mid rate for residential customers and evening rate for commercial customers).
- If the customer is on a tariff with two or more price steps, the customer's network energy tariff rate for the purpose of paying the customer's retailer shall be the first rate applied.
- The customer will continue to pay the network access charge.
- This arrangement is available only to customers with less than 30 kilowatts installed capacity of renewable generation.

In addition, ActewAGL Distribution is in the process of making time-of-use network tariffs the default tariffs for new premises. ActewAGL Distribution would support the application of time-of-use tariffs to new customers moving into premises with time-of-use meters installed.

Table 3.9 compares the proposed network charges in 2010/11 with those in 2009/10. The average change in network charges is shown in cents per kWh and as a percentage for an average consumer for each tariff.²⁵

Table 3.9 Changes in network charges

Code	Description	Unit	Network Charges 2009/10	Network Charges 2010/11	Average change c/kWh	Average change %
Residential tariffs						
010	Residential Basic Network				0.453	7.4%
	Network access charge	cents/day	12.97	14.01		
	Energy consumption	cents/kWh	5.45	5.85		
015	Residential TOU Network				0.366	6.4%
	Network access charge	cents/day	12.97	14.01		
	Energy consumption at max times	cents/kWh	7.87	8.43		
	Energy consumption at mid times	cents/kWh	4.59	4.75		
	Energy consumption at economy times	cents/kWh	3.24	3.53		
020	Residential 5000 Network				0.449	8.4%
	Network access charge	cents/day	34.37	35.51		
	Energy consumption for the first 60 kWh per day	cents/kWh	3.85	4.25		
	Energy consumption above 60 kWh per day	cents/kWh	5.45	5.85		

²⁴ As established by the *Electricity Feed-in (Renewable Energy Premium) Act 2008* (Feed-in tariff Act)

²⁵ The average change in network charges is calculated by determining the average revenue for each tariff using the prices for each year and taking the difference. The percentage change is this difference divided by the average revenue with 2008/09 prices. Where there are no customers, it is not possible to calculate the average price.

030 Residential with Heat Pump Network				0.415	10.5%
Network access charge	cents/day	77.27	78.51		
Energy consumption for the first 165 kWh per day	cents/kWh	2.35	2.74		
Energy consumption above 165 kWh per day	cents/kWh	5.45	5.85		
060 Off-Peak (1) Night Network				0.380	48.7%
Energy consumption	cents/kWh	0.78	1.16		
070 Off-Peak (3) Day & Night Network				0.240	18.2%
Energy consumption	cents/kWh	1.32	1.56		
Renewable Energy Generation					
Energy consumption	cents/kWh	0.00	0.00		
Commercial low voltage					
040 General Network				0.795	9.0%
Network access charge	cents/day	26.00	27.66		
Energy consumption for the first 330 kWh per day	cents/kWh	8.27	9.03		
Energy consumption above 330 kWh per day	cents/kWh	10.99	11.90		
135 Small Unmetered Loads Network				0.269	2.8%
Network access charge	cents/day	18.80	20.30		
Energy consumption	cents/kWh	9.53	9.79		
141 Internal Network (Obsolete)				8.265	568%
Network access charge	cents/day	2.40	27.66		
Energy consumption	cents/kWh	1.395	9.03		
080 Streetlighting Network				0.270	5.3%
Network access charge	cents/day	26.00	28.00		
Energy consumption	cents/kWh	5.10	5.37		
090 General TOU Network				0.520	6.9%
Network access charge	cents/day	26.00	27.66		
Energy consumption at business times	cents/kWh	12.93	14.00		
Energy consumption at evening times	cents/kWh	7.12	7.32		
Energy consumption at off-peak times	cents/kWh	2.94	3.08		
Low voltage time of use demand network					
101 LV TOU kVA Demand Network				0.520	8.6%
Network access charge per connection point	cents/day	32.00	35.00		
Maximum demand charge	c/kVA/day	33.20	37.60		
Energy consumption at business times	cents/kWh	4.21	4.42		
Energy consumption at evening times	cents/kWh	3.07	2.97		
Energy consumption at off-peak times	cents/kWh	1.58	1.64		
103 LV TOU Capacity Network				0.374	8.2%
Network access charge per connection point	cents/day	32.00	35.00		
Maximum demand charge	c/kVA/day	15.00	17.30		
Capacity charge	c/kVA/day	15.00	17.30		
Energy consumption at business times	cents/kWh	4.21	4.42		
Energy consumption at evening times	cents/kWh	3.07	2.97		
Energy consumption at off-peak times	cents/kWh	1.58	1.64		

High voltage tariffs

High voltage time of use demand network with ActewAGL low voltage network

111 HV TOU Demand Network				0.442	8.5%
Network access charge per connection point	\$/day	12.00	13.00		
Maximum demand charge	c/kVA/day	13.00	15.20		
Capacity charge	c/kVA/day	13.00	15.20		
Energy consumption at business times	cents/kWh	3.90	3.95		
Energy consumption at evening times	cents/kWh	2.81	2.63		
Energy consumption at off-peak times	cents/kWh	1.48	1.50		
112 HV TOU Demand Network – Customer HV					
Network access charge per connection point	\$/day	12.00	13.00		
Maximum demand charge	c/kVA/day	12.00	14.20		
Capacity charge	c/kVA/day	12.00	14.20		
Energy consumption at business times	cents/kWh	3.90	3.95		
Energy consumption at evening times	cents/kWh	2.81	2.63		
Energy consumption at off-peak times	cents/kWh	1.48	1.50		

High voltage time of use demand network without ActewAGL low voltage network

121 HV TOU Demand Network – Customer LV				0.354	8.2%
Network access charge per connection point	\$/day	12.00	13.00		
Maximum demand charge	c/kVA/day	13.00	15.20		
Capacity charge	c/kVA/day	13.00	15.20		
Energy consumption at business times	cents/kWh	3.47	3.53		
Energy consumption at evening times	cents/kWh	2.43	2.26		
Energy consumption at off-peak times	cents/kWh	1.31	1.34		
122 HV TOU Demand Network – Customer HV and LV					
Network access charge per connection point	\$/day	12.00	13.00		
Maximum demand charge	c/kVA/day	12.00	14.20		
Capacity charge	c/kVA/day	12.00	14.20		
Energy consumption at business times	cents/kWh	3.47	3.53		
Energy consumption at evening times	cents/kWh	2.43	2.26		
Energy consumption at off-peak times	cents/kWh	1.31	1.34		

The increases shown in the table reflect the increases in both DUOS and TUOS charges for 2010/11. The CPI of 1.82 per cent and the X factor of 4.0 per cent raise the MAAR by 5.89 per cent. In addition, there is a 22.3 per cent increase in average TUOS charges in 2010/11. The average increase in NUOS charges is 0.49 c/kWh or 8.4 per cent.

Table 3.10 compares the proposed miscellaneous fees and charges for 2010/11 with the comparable charges for 2009/10. The average increase in these miscellaneous fees and charges is around 3 per cent.

Table 3.10 Changes in miscellaneous services charges

		Proposed			
Code	Description	Unit	Prices 2009/10	Prices 2010/11	Change %
For visit to re-energise or de-energise a site					
501	Business hours – re-energise a site	per visit	\$47.27	\$49.09	3.9%
502	After hours – re-energise a site	per visit	\$102.73	\$105.45	2.6%
503	Business hours – de-energise a site	per visit	\$42.73	\$43.64	2.1%
505	De-energise a site for non-payment	per visit	\$79.09	\$81.82	3.5%
508	Field visit only (De-energise a site for non-payment)	per visit	\$57.27	\$59.09	3.2%
Temporary connections					
520	Overhead	per installation	\$340.00	\$350.00	2.9%
521	Standard underground	per installation	\$493.64	\$508.18	2.9%
522	Free-standing underground	per installation	\$598.18	\$616.36	3.0%
Services					
530	Overhead: remove, reposition or disconnect service	per site visit	\$244.55	\$251.82	3.0%
531	Underground: remove, reposition or disconnect service	per site visit	\$589.09	\$606.36	2.9%
Upgrade service from single to three phase at customer's request where load does not justify three phase					
532	Overhead	per installation	\$316.36	\$325.45	2.9%
533	Underground – service cable replacement not required	per installation	\$316.36	\$325.45	2.9%
534	Underground – service cable replacement required	per installation	\$589.09	\$606.36	2.9%
540	Rescheduled visit (applied where a revisit to a site is necessitated by obstructed access, non-compliance with the Service and Installation Rules or the client is not ready for the scheduled work.)				
550	Issue of copies of electrical drawings	per sheet or electronic equivalent	\$162.73	\$167.27	2.8%
560	De-energising wires (to allow safe approach, for example, for tree pruning, plant operation, oversize loads, construction activities)	per occurrence	\$392.73	\$404.55	3.0%
570	Embedded generation - operational and maintenance charges for small embedded generators (other renewable <30KW)				
	for connection assets		2.0%	2.0%	0.0%
	for shared network assets (if applicable)		2.0%	2.0%	0.0%

The “Installation defect” charge, which applies where a second visit to a site was necessitated by obstructed access or non-compliance with the Service and Installation Rules, is being renamed the “Rescheduled visit” charge. Also, the application of this charge is being

extended from 1 July 2010 to include any situation where the client is not ready for scheduled work. In 2009, it was not necessary to apply this charge as no event occurred. It is not designed to generate revenue. It is intended to reflect the cost of a site visit and encourage clients to advise ActewAGL Distribution if they are not ready for a planned visit, thereby reducing costs and improving productivity.

4. Charges for metering services

4.1 The structure and basis of ActewAGL Distribution's metering charges

ActewAGL Distribution's schedule of metering charges comprises daily charges for the provision of each of 4 different types of manually read metering services and charges for special meter reads. Meters for residential consumers are usually read quarterly and meters for small low voltage commercial consumers (consuming less than 160 MWh pa) are usually read monthly. Metering services for larger low voltage consumers and for high voltage consumers are open to competition and are not regulated.

The metering charges for small consumers are set on a cost reflective basis. The costs of meter reading and meters (including the costs of depreciation, return on capital, installation costs and maintenance and administration costs) have been calculated for each meter type. The costs are applied to the number of meters of each type to derive the total cost for each meter type. The meter types are each allocated to a class of metering service and the total cost of each class is allocated to the customers in that class to estimate the average cost of the metering service to each class.

4.2 Proposed metering services charges for 2010/11

Prices for ActewAGL Distribution's alternative control services are regulated using a maximum revenue cap.²⁶ This cap is applied to metering revenue that would have been earned from the proposed prices for the coming financial year if they had been applied to the quantities sold in the services provided in the previous calendar year.

In 2008/09, the revenue cap for alternative control services was \$5,666,539 before adjustments. In the first year of the 2009–14 regulatory period, this cap was set using a CPI - X adjustment. In subsequent years, the cap is increased by CPI. Table 4.1 shows the X factor for 2009/10 and CPI factors applied to calculate the revenue cap for alternative control services. The approach is similar to that applied in Table 3.1 to calculate the MAAR.

Table 4.1 Alternative control services revenue cap

	<i>Alternative services revenue in previous year</i>	<i>X Factor</i>	<i>Sum of CPI indices</i>	<i>CPI</i>	<i>Alternative service revenue cap</i>
2008/09			631.8		\$5,666,539
2009/10	\$5,666,539	-29.30%	659.3	4.35%	\$7,645,746
2010/11	\$7,645,746	0.00%	671.3	1.82%	\$7,784,907

As shown in Table 4.1, the alternative control services revenue cap for 2010/11 is \$7,784,907.

²⁶ AER 2009, *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision*, April, p. 146

To demonstrate compliance, ActewAGL Distribution has prepared the required schedule of metering charges showing the revenue from each charge if that charge had applied in the 2009 calendar year.²⁷ The schedule shows that the revenue from these charges does not exceed the revenue cap for alternative control services. The proposed metering charges and the forecast revenue from those charges are shown in Table 4.2.

Table 4.2 Proposed metering charges 2010/11

Code	Description	Unit	Units 2009	Prices excl. GST 2010/11	Revenue
MP1	Quarterly basic metering rate Accumulation and time-of-use meters read quarterly	- cents per day per NMI	140,073	12.79	\$6,539,096
MP2	Monthly basic metering rate Accumulation and time-of-use meters read monthly	- cents per day per NMI	11,817	22.34	\$963,551
MP3	Time-of-use metering rate Time-of-use meters read monthly	- cents per day per NMI	139	22.34	\$11,373
MP4	Monthly manually-read interval metering rate Interval meters recording at either 15 or 30 minute intervals, read manually and processed monthly	\$ per day per NMI	361	\$1.82	\$239,812
MP5	Internal metering rate Sites entitled to the Internal Network charge	- cents per day per NMI	108	22.34	\$8,806
MP6	Quarterly manually-read interval metering rate Interval meters recording at either 15 or 30 minute intervals, read manually and processed quarterly	cents per day per NMI	0	52.50	\$0
504	Meter test during business hours (deposit)	per test	37	\$60.91	\$2,270
506	Special meter reading, for example, check read (second in a billing period)	per read	21	\$31.82	\$668
507	Install interval meter at customer's request	per meter	0	\$152.73	\$0
509	Replace meter to facilitate renewable energy installation	per meter	126	\$152.73	\$19,237
Total			152,682		\$7,784,813

*national meter identifier.

The revenue from the 2009 units of \$7,784,813 is less than the revenue cap for alternative control services of \$7,784,907. Therefore, the charges comply with the AER's final decision on alternative control services.

²⁷ AER 2009, *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision*, April, p. 153.

4.3 Proposed changes in metering charges

There are no changes to metering services or to the structure of metering charges in 2010/11.

Table 4.3 compares the metering charges for 2010/11 with those for 2009/10.

Table 4.3 Changes in metering charges

Code	Description	Unit	Prices 2009/10	Proposed prices 2010/11	Change %
MP1	Quarterly basic metering rate Accumulation and time-of-use meters read quarterly	- cents per day per NMI	12.23	12.79	4.6%
MP2	Monthly basic metering rate Accumulation and time-of-use meters read monthly	- cents per day per NMI	21.40	22.34	4.4%
MP3	Time-of-use metering rate Time-of-use meters read monthly	- cents per day per NMI	21.40	22.34	4.4%
MP4	Monthly manually-read interval metering rate Interval meters recording at either 15- or 30-minute intervals, read manually and processed monthly	\$ per day per NMI	\$1.75	\$1.82	4.0%
MP5	Internal metering rate Sites entitled to the Internal Network charge	- cents per day per NMI	7.50	22.34	198%
MP6	Quarterly manually-read interval metering rate Interval meters recording at either 15- or 30-minute intervals, read manually and processed quarterly	cents per day per NMI	48.00	52.50	9.4%
504	Meter test during business hours (deposit)	Per test	\$59.09	\$60.91	3.1%
506	Special meter reading , for example, check read (second in a billing period)	per read	\$30.91	\$31.82	2.9%
507	Install interval meter at customer's request	per meter	\$147.27	\$152.73	3.7%
509	Replace meter to facilitate renewable energy installation	per meter	\$147.27	\$152.73	3.7%

5. Indicative customer impacts

5.1 Proposed changes in network and metering charges

ActewAGL Distribution's customers who consume less than 160 MWh per annum are most likely to be subject to charges for both standard control services (network tariffs) and alternative control services (regulated metering service charges). Table 5.1 shows network charges (standard control services and TUOS) plus metering service charges (alternative control services) for 2010/11 and the comparable charges for 2009/10, excluding GST. High voltage charges have been excluded as ActewAGL Distribution does not provide metering services to high voltage customers. Metering services to customers consuming more than 160 MWh per annum are open to competition and not regulated.

Standard control services and alternative control services are shown separately in ActewAGL Distribution's pricing schedule and are charged separately to retailers.

Table 5.1 Network and metering charges 2010/11

Code	Description	Unit	Network & metering charges 2009/10	Network & metering charges 2010/11	Average change c/kWh	Average change %
Residential tariffs						
010	Residential Basic Network				0.48	7.2%
	Network access charge	cents/day	25.20	26.80		
	Energy consumption	cents/kWh	5.45	5.85		
015	Residential TOU Network				0.39	6.2%
	Network access charge	cents/day	25.20	26.80		
	Energy consumption at max times	cents/kWh	7.87	8.43		
	Energy consumption at mid times	cents/kWh	4.59	4.75		
	Energy consumption at economy times	cents/kWh	3.24	3.53		
020	Residential 5000 Network				0.47	8.0%
	Network access charge	cents/day	46.60	48.30		
	Energy consumption for the first 60 kWh per day	cents/kWh	3.85	4.25		
	Energy consumption above 60 kWh per day	cents/kWh	5.45	5.85		
030	Residential with Heat Pump Network				0.43	10.2%
	Network access charge	cents/day	89.50	91.30		
	Energy consumption for the first 165 kWh per day	cents/kWh	2.35	2.74		
	Energy consumption above 165 kWh per day	cents/kWh	5.45	5.85		
060	Off-Peak (1) Night Network					48.7%

					0.38	
	Energy consumption	cents/kWh	0.78	1.16		
070	Off-Peak (3) Day & Night Network				0.24	18.2%
	Energy consumption	cents/kWh	1.32	1.56		
	Renewable Energy Generation					
	Energy consumption	cents/kWh	0.00	0.00		
Commercial low voltage tariffs						
040	General Network				0.81	8.8%
	Network access charge	cents/day	47.40	50.00		
	Energy consumption for the first 330 kWh per day	cents/kWh	8.27	9.03		
	Energy consumption above 330 kWh per day	cents/kWh	10.99	11.90		
135	Small Unmetered Loads Network				0.27	2.8%
	Network access charge	cents/day	18.80	20.30		
	Energy consumption	cents/kWh	9.53	9.79		
080	Streetlighting Network				0.27	5.3%
	Network access charge	cents/day	47.40	50.34		
	Energy consumption	cents/kWh	5.10	5.37		
090	General TOU Network				0.52	6.9%
	Network access charge	cents/day	47.40	50.00		
	Energy consumption at business times	cents/kWh	12.93	14.00		
	Energy consumption at evening times	cents/kWh	7.12	7.32		
	Energy consumption at off-peak times	cents/kWh	2.94	3.08		
Low voltage time of use demand network						
101	LV TOU kVA Demand Network				0.52	8.5%
	Network access charge per connection point	cents/day	207	217		
	Maximum demand charge	c/kVA/day	33.20	37.60		
	Energy consumption at business times	cents/kWh	4.21	4.42		
	Energy consumption at evening times	cents/kWh	3.07	2.97		
	Energy consumption at off-peak times	cents/kWh	1.58	1.64		
103	LV TOU Capacity Network				0.38	8.2%
	Network access charge per connection point	cents/day	207	217		
	Maximum demand charge	c/kVA/day	15.00	17.30		
	Capacity charge	c/kVA/day	15.00	17.30		
	Energy consumption at business times	cents/kWh	4.21	4.42		
	Energy consumption at evening times	cents/kWh	3.07	2.97		
	Energy consumption at off-peak times	cents/kWh	1.58	1.64		

5.2 Estimated impacts on average customer electricity bills

The proposed 2010/11 increases in network and metering charges would raise the electricity bill for an average residential customer consuming 7000 kWh on the Residential Basic network charge by \$0.72 per week (including GST), a real increase of 5.2 per cent (7.1 per cent nominal). For a commercial customer consuming 30 MWh per annum on General network charge, the proposed network and metering price increases would raise their electricity bill by \$5.02 per week (including GST) or 7.0 per cent real (8.9 per cent nominal).

Attachment 1: Compliance with transitional Rules requirements

Table A1.1 provides a checklist of where the relevant requirements of Part I of the transitional *Rules* are met in this pricing proposal. One of the transitional *Rules* requirements is that the pricing proposal demonstrates compliance with any applicable distribution determination (clause 6.18.2 (7)). Table A1.2 provides a separate checklist of the relevant requirements from the AER's *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision* and where they are addressed in this pricing proposal.

Table A1.1: Checklist of transitional Rules pricing proposal requirements

Rules	Requirement	Coverage in this pricing proposal
6.18.2 (b)	The pricing proposal must: (1) set out the tariff classes that are to apply for the relevant regulatory year; and (2) set out the proposed tariffs for each tariff class; and (3) set out, for each proposed tariff, the charging parameters and the elements of service to which each charging parameter relates; and (4) set out, for each tariff class related to standard control services, the expected weighted average revenue for the relevant regulatory year and also for the current regulatory year; and (5) set out the nature of any variation or adjustment to the tariff that could occur during the course of the regulatory year and the basis on which it could occur; and (6) set out how charges incurred by the Distribution Network Service Provider for transmission use of system services are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those charges in the previous regulatory year; and (7) demonstrate compliance with the Rules and any applicable distribution determination; and (8) describe the nature and extent of change from the previous regulatory year and demonstrate that the changes comply with the Rules and any applicable distribution determination.	(1) The tariff classes for standard control services are set out in Section 2.1. (2) Proposed distribution tariffs are in Table 3.3. Proposed metering charges are in Table 4.2. (3) Section 2.1 and Tables 2.1, 2.2 and 2.3 set out each charging parameter and the element of service to which it relates. (4) Table 3.4 shows the weighted average DOUS revenue for 2009/10 and 2010/11. (5) ActewAGL Distribution does not propose any variations or adjustments to tariffs for 2010/11. However, any variations would be made in accordance with the cost pass through mechanisms in the AER's final determination. (6) Section 3.4 provides an explanation of how TUOS charges are passed on to customers, and ActewAGL Distribution's proposed adjustment for under recovery of TUOS costs in 2010/11. Proposed TUOS charges are provided in Table 3.6. (7) Table A1.2 below provides the checklist for relevant requirements from the 2009-14 distribution determination. (8) Sections 3.6 and 4.3 contain descriptions of the nature and extent of changes from the 2009/10 regulatory year. Table 3.3 and Table 3.5 demonstrate that revenue from 2010/11 prices matches allowable revenue calculated in section 3.1.1
6.18.3	(a) A pricing proposal must define the tariff classes into which customers for direct control services are divided. (b) Each customer for direct control services must be a member of 1 or more tariff classes. (c) Separate tariff classes must be constituted for	(a) Section 2.1 provides an explanation of the tariff classes for standard control services. Section 4.1 provides an explanation of the categories of charges for alternative control services. (b) Each customer is on one or more tariffs within one or more tariff classes for which qualify. (c) Separate tariff classes and charges are specified

	customers to whom standard control services are supplied and customers to whom alternative control services are supplied (but a customer for both standard control services and alternative control services may be a member of 2 or more tariff classes). (d) A tariff class must be constituted with regard to: (a) the need to group customers together on an economically efficient basis; and (b) the need to avoid unnecessary transaction costs.	for standard control services in Table 3.3 and Table 3.5 and alternative control services in Table 4.2. (d) Section 2.1 contains an explanation of the basis of the tariff classes.
6.18.4	(a) In formulating provisions of a distribution determination governing the assignment of customers to tariff classes or the re-assignment of customers from one tariff class to another, the AER must have regard to the following principles: (1) customers should be assigned to tariff classes on the basis of one or more of the following factors: (i) the nature and extent of their usage; (ii) the nature of their connection to the network; (iii) whether remotely-read interval metering or other similar metering technology has been installed at the customer's premises as a result of a regulatory obligation or requirement; (2) customers with a similar connection and usage profile should be treated on an equal basis; (3) however, customers with micro-generation facilities should be treated no less favourably than customers without such facilities but with a similar load profile; (4) a Distribution Network Service Provider's decision to assign a customer to a particular tariff class, or to re-assign a customer from one tariff class to another should be subject to an effective system of assessment and review.	(a) Appendix A of the AER's final decision sets out the procedures for assigning or re-assigning customers to tariff classes.
6.18.5	(a) For each tariff class, the revenue expected to be recovered should lie on or between: (1) an upper bound representing the stand alone cost of serving the customers who belong to that class; and (2) a lower bound representing the avoidable cost of not serving those customers. (b) A tariff, and if it consists of 2 or more charging parameters, each charging parameter for a tariff class: (1) must take into account the long run marginal cost for the service or, in the case of a charging parameter, for the element of the service to which the charging parameter	(a) Table 2.4 shows that the revenue expected from each tariff class lies between the stand alone cost and the avoidable cost for the tariff class. Section 2.3 contains an explanation of the basis and application of the principle. (b) Section 2.3 contains explanations of how ActewAGL Distribution takes into account long run marginal costs, transactions costs, the scope for customers to respond to price signals and the need to recover revenue in an efficient manner (as required by (c) below).

	relates; and (2) must be determined having regard to: (i) transaction costs associated with the tariff or each charging parameter; and (ii) whether customers of the relevant tariff class are able or likely to respond to price signals. (c) If, however, as a result of the operation of paragraph (b), the Distribution Network Service Provider may not recover the expected revenue, the provider must adjust its tariffs so as to ensure recovery of expected revenue with minimum distortion to efficient patterns of consumption.	(c) Section 2.3 contains explanations of how the need to recover revenue with minimum distortion to efficient patterns of consumption is taken into account when setting charging parameters.
6.18.6	(a) This clause applies only to <i>tariff classes</i> related to the provision of <i>standard control services</i>. (b) The expected weighted average revenue to be raised from a <i>tariff class</i> for a particular <i>regulatory year</i> of a <i>regulatory control period</i> must not exceed the corresponding expected weighted average revenue for the preceding <i>regulatory year</i> by more than the permissible percentage. (c) The permissible percentage is the greater of the following: (1) the CPI-X limitation on any increase in the <i>Distribution Network Service Provider's</i> expected weighted average revenue between the two <i>regulatory years</i> plus 2%; Note: The calculation is of the form $(1 + \text{CPI})(1 - X)(1 + 2\%)$ (2) CPI plus 2%. Note: The calculation is of the form $(1 + \text{CPI})(1 + 2\%)$ (d) In deciding whether the permissible percentage has been exceeded in a particular <i>regulatory year</i>, the following are to be disregarded: (1) the recovery of revenue to accommodate a variation to the distribution determination under rule 6.6 or 6.13; (2) the recovery of revenue to accommodate pass through of charges for transmission use of system services to customers. (e) This clause does not, however, limit the extent a tariff for customers with remotely-read interval metering or other similar metering technology may vary according to the time or other circumstances of the customer's usage.	Section 3.2 and Table 3.4 show that tariff classes comply with the side constraints.
6.18.7	(a) A <i>pricing proposal</i> must provide for tariffs designed to pass on to customers the charges to be incurred by the <i>Distribution Network Service Provider</i> for <i>transmission use of system services</i>. (b) The amount to be passed on to customers for a particular <i>regulatory year</i> must not exceed the estimated amount of the <i>transmission use of system</i> charges for the relevant <i>regulatory year</i> adjusted for over or under	Section 3.4 provides an explanation of how TUOS charges are passed on to customers, and ActewAGL Distribution's proposed adjustment for under recovery of TUOS costs in 2009/10. Proposed TUOS charges are provided in Table 3.7.

	recovery in the previous <i>regulatory year</i>. (c) The extent of the over or under recovery is the difference between: (1) the amount actually paid by the <i>Distribution Network Service Provider</i> by way of <i>transmission use of system charges</i> in the previous <i>regulatory year</i>; and (2) the amount passed on to customers by way of <i>transmission use of system charges</i> by the <i>Distribution Network Service Provider</i> in the previous <i>regulatory year</i>.	
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Table A1.2: Checklist of requirements from the AER final decision

<i>Final decision requirement</i>	<i>Coverage in this pricing proposal</i>
<i>Compliance with X factors</i> "In accordance with clause 6.12.1(11) of the transitional chapter 6 rules the X factors to apply to ActewAGL are set out in table 17.8 of the final decision" (AER Final Decision, p. 145)	Table 3.1 shows that the X factor applied to the calculation of MAAR for standard control services is 4.0% and therefore complies with the AER's final decision as set out in table 17.8 of the final decision. Section 4.2 demonstrates that an X factor of 0% has been applied in calculating the revenue cap for alternative control services. This complies with the AER's Final Decision, as set out in table 17.8.
<i>Compliance with the standard control services control mechanism.</i> "In accordance with clause 6.12.1(13) of the transitional chapter 6 rules compliance with the maximum allowable average revenue cap for standard control services must be demonstrated by ActewAGL using the formulae outlined in section 4.5 and in accordance with appendix F of this final decision." (AER Final Decision, p. 20)	Section 3.1, shows ActewAGL Distribution's application of the formulae set out in section 4.5 of the AER final decision to derive the revenue cap for standard control services. Section 3.2 and Table 3.3 demonstrate that the revenue from standard control services derived from 2010/11 tariffs complies with this cap. No adjustments under Appendix F are proposed. The change to the time-of-use definitions (excluding public holidays) for commercial customers will apply as meters are replaced or reprogrammed. This will occur gradually over 10 years. The changes to the load profile for pricing will occur as the data changes. In addition, time-of-use prices are in the process of being made the default tariff for new premises and this data will feed into the pricing load profile over time.
<i>Compliance with the alternative control services control mechanism.</i> "In accordance with clause 6.12.1(13) of the transitional chapter 6 rules ActewAGL must demonstrate compliance with the control mechanism for alternative control services by submitting to the AER a schedule of metering charges, in the form of table 13.5 of ActewAGL's revenue proposal, as soon as practicable after prices for each regulatory year are determined." (AER Final Decision, p. 123)	Section 4.2 contains Table 4.2 which is in the same format as Table 13.5 in ActewAGL Distribution's regulatory proposal.
<i>TUOS unders and overs account.</i> "In accordance with clause 6.12.1(19) of the transitional chapter 6 rules ActewAGL must submit, as part of its annual pricing proposal, a record of the amount of revenues recovered from TUOS charges and associated payments in accordance with appendix E of this final decision." (AER	The TUOS unders and overs account is shown in section 3.4 (Table 3.6).

Final Decision, p. 20)	
<i>Side constraints</i> "In accordance with clause 6.18.2(b)(7) and (8) of the transitional chapter 6 rules ActewAGL must ensure its annual pricing proposals comply with the side constraint formula defined in section 4.5 of this final decision." (AER final decision, p. 20)	Side constraints are considered in section 3.2 and compliance is shown in Table 3.4.

Attachment 2: Analysis of ActewAGL's electricity distribution services costs, NERA report

Commercial-in-confidence report supplied separately.